Supplementary Material for "Statistical Utility of Cox Regression Under Differential-Privacy-Style Perturbation: Clipping Bounds, Local and Central Mechanisms, and the Price of Formal Guarantees"

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Figures are numbered S1–S8 and tables S1–S10, continuously within this document. Two items that would otherwise appear here are printed in the main manuscript itself: the variable-type breakdown as Table I, and the formal-guarantee comparison as Table VI. Everything the main manuscript refers to with an S number is here.

Contents. Section S0 presents the methodological details that the page limit keeps out of the main manuscript (preprocessing, budget allocation, clipping bounds and the time axis, the two simulation series and their seeds, and the normalization of the comparison arm of Section IV-I). Section S1 lists the disclosures concerning the fixed-bound protocol. Section S2 presents the full utility results underlying Figs. 1–3 of the main manuscript (Tables S1–S3, Fig. S1). Section S3 provides the false-positive analyses on the inverted-U pattern (Figs. S2–S5, Table S4). Section S4 presents the mechanism-selection and discrete-time diagnostics and the self-referential Phase 3 baseline (Tables S5–S7). Section S5 presents the clipping-bound comparison, the fixed bounds with their sources, the thresholds under the three strategies, and the formal comparison arm (Tables S8–S10, Fig. S6). Section S6 presents the hazard-ratio results (Figs. S7–S8).

Unless stated otherwise, every result uses $B = 1{,}000$ Monte Carlo iterations, the 15-point ε grid of the main manuscript, and the data-driven clipping bounds of the main analysis.

S0. METHODS DETAILS

Preprocessing. Each dataset was taken from the R survival package and reduced as follows; all times are in days.

- **lung**: 228 → 168 rows, complete cases on the nine variables used (60 excluded); event is death; no imputation.
- **pbc**: 418 → 312 rows, randomized patients only (106 with missing `trt` excluded); event is death, transplant censored; median imputation of `chol` (28 values), `copper` (2), `trig` (30), `platelet` (4).
- **colon**: 1,858 → 929 rows, the recurrence records only (`etype` = 1); event is recurrence; median imputation of `nodes` (18) and mode imputation of `differ` (23).
- **rotterdam**: 2,982 rows, all used, with `rtime`/`recur` as the outcome; event is recurrence; no imputation.
- **flchain**: 7,874 → 6,524 rows, complete cases on `creatinine` (1,350 excluded), `chapter` and `flc.grp` dropped; event is death; no imputation.

Imputed values are perturbed like observed values and enter the observed ranges and the baseline fits. Ordinal variables coded numerically in the package (pbc `edema`, `stage`; colon `differ`, `extent`) are perturbed by k-ary Randomized Response over their coded levels and enter the Cox model as single numeric terms; nominal factors (colon `rx`, rotterdam `size`) are dummy coded. A categorical variable with $k = 2$ is perturbed by categorical Randomized Response, which coincides with binary Randomized Response.

Budget allocation. Input perturbation splits ε over the q variables ($\varepsilon_j = \varepsilon/q$ in Phase 1, $\varepsilon/(q+2)$ in Phases 2–3), every variable receiving the same ε_j whatever its type. Output perturbation splits ε over the p Cox coefficients ($\varepsilon_j = \varepsilon/p$, sequential composition; p counts coefficients after dummy coding), and coefficient j receives an independent draw from $\mathrm{Lap}(p\,\Delta\hat{\beta}_j/\varepsilon)$ with $\Delta\hat{\beta}_j = \max_i |\mathrm{dfbeta}_{i,j}|$ taken column-wise; in the 70/30 series $\Delta\hat{\beta}_j$ is recomputed on each training split.

Clipping bounds and the time axis. The data-driven bounds $[\min_j, \max_j]$ and the Phase 2 upper limit $T_{\max}$ are the observed extremes of the full prepared dataset, held as constants in both simulation series (so the held-out rows of the 70/30 series inform them; no guarantee is claimed for these arms). For flchain alone, $T_{\max}$ is the maximum follow-up of the source data before the complete-case exclusion (5,215 days; the prepared dataset reaches 5,166); therefore, its Phase 2 noise scale is 0.95% larger than what the prepared extreme would provide; the results are reported as run. The Phase 2 noise on T has scale $T_{\max}/\varepsilon_j$ and the result is clamped to $[0, T_{\max}]$. In Phase 3 the interval boundaries are quantiles of the observed event

times—a data-dependent step charged no budget—and the perturbation of (k_i^*, δ_i) and of the covariates precedes the expansion into person-period rows, which is post-processing of the perturbed record.

Two simulation series. LSR, FPR and hazard-ratio bias come from fits to the full dataset (in-sample); the test C-index and ΔC come from a separate series of $B = 1{,}000$ stratified 70/30 splits. In every arm, the model is fitted to the training split (perturbed, in the input-perturbation arms), its training C-index is computed on that same split, and the test C-index scores the unperturbed test split, that is, the released model applied to clean records. For Phase 3, the risk score of the test C-index is the covariate linear predictor of the logistic model (spline and interval terms excluded) applied to the unperturbed test covariates, scored against the continuous-time outcome. The training C-index likewise scores the perturbed training covariates, with the covariate coefficients only, against the unperturbed training outcome at the individual level; in Phase 2 alone the training outcome is itself perturbed. Seeds are base 42 with dataset offsets 0, 10^5, 2×10^5, 3×10^5, 4×10^5 (lung, pbc, colon, rotterdam, flchain) and per-run seed $\text{base} + (\text{index of } \varepsilon - 1) \times B + \text{iteration}$; split seeds are $10^5 + \text{iteration}$; the fixed and widened bound arms add 10^6 and 2×10^6; the comparison arm uses base 3,100,000 (four specifications, seeds 3,100,001–3,400,000). Every result row records its seed.

Normalization in the comparison arm of Fig. S6. In the fully data-independent specification, each covariate is centered at the midpoint of its box from the source protocol, divided by the half-width, clipped to $[-1, 1]$ and scaled by $1/\sqrt{p}$, so that the box lies inside the unit sphere the algorithm assumes; time is divided by the fixed horizon (5,223 days: the interval from the start of enrolment on 13 March 1995 to the end of follow-up on 30 June 2009 published for the source cohort by Dispenzieri et al. (2012). The two data-dependent specifications center each covariate at its training-column mean and divide all columns by the largest training row norm, as the original description implies; the customary one also divides time by the largest training follow-up. The fourth specification, fully data-independent with 13 equal-width intervals, uses the same box, scaling, and horizon as the fully data-independent 200-interval one and differs from it only in the interval count.

S1. Disclosures Concerning the Fixed-Bound Protocol

The fixed-bound protocol was specified after the datasets were available (Section III-I of the main manuscript). The complete list of eight disclosures follows, with the reasoning for each.

1) **The direction of the `copper` replacement is outward.** Obtaining the full text of its source widened the bound from 2,000 to 4,000 µg/24 h, doubling the Laplace noise scale; hence, the fixed arm performs *worse* than before the correction. The replacement was made for conformity with the protocol, not for performance, but the direction is disclosed since it removes any suspicion that bounds were selected to flatter the results.
2) **One clause was written for a single variable and now applies to none.** It was added when only the abstract of that variable's source was available, and the observed range could not be read from it. The full text supplied the range, a higher-priority clause applied instead, and the added clause is now unused. We retain it in the protocol; hence, the episode remains auditable. There is no mitigating circumstance: it was a new clause written after seeing the values it governed.
3) **Widening that bound weakened a summary computed during the work.** The count of covariates whose fixed-to-observed width ratio is below 2.0 returned from 17 to 16 of 27; hence, the finding that formalization costs less than a doubling of noise for most variables is one variable weaker than when first computed.
4) **For the free light chain covariates, we did not adopt the observed range of the source series.** Its individual values are published only as a figure; hence, the range is not re-derivable in the form the protocol requires. The basis is that the range is not available in a reconstructable form, not that it does not exist; had it been published numerically, the bound would have been wider than the one we adopted.
5) **The first four clauses were all written after the 12 laboratory variables had been inspected.** The defense against arbitrariness is therefore the full disclosure of rejected candidates rather than the priority of the rules.
6) **The clause covering covariates whose definition names a subpopulation was also written after the fact**, and it applies to only two variables. The bound it yields rests on a reference interval whose source population contains a subgroup reaching four times that bound: 3.5% of an unselected hemodialysis population is monoclonal, with κ reaching 122.4 mg/dL against our bound of 30. Such a value in our sample would be clipped, and the local-DP guarantee would still hold.
7) **One clause was recast during the work**, from a test based on terminology to one based on the structure of the source's reporting. The recasting was also conducted after the 12 variables had been seen, though the replacement is the less arbitrary of the two because it turns on how a source is written rather than on what value it reports. No adopted bound changed.
8) **Two sentences were added to the highest-priority clause**, specifying which of several reported groups to take and how to treat an upper limit resting on a single case. These were written after seeing the values they govern, and no adopted bound changed. They exist because without them a third party could not re-derive the same bound from the protocol alone, which is the property the protocol claims.

S2. Full Utility Results

Tables S1–S3 present the numerical values summarized by Figs. 1–3 in the main manuscript. Fig. S1 illustrates the per-coefficient decomposition of loss of significance, which did not fit the main manuscript.

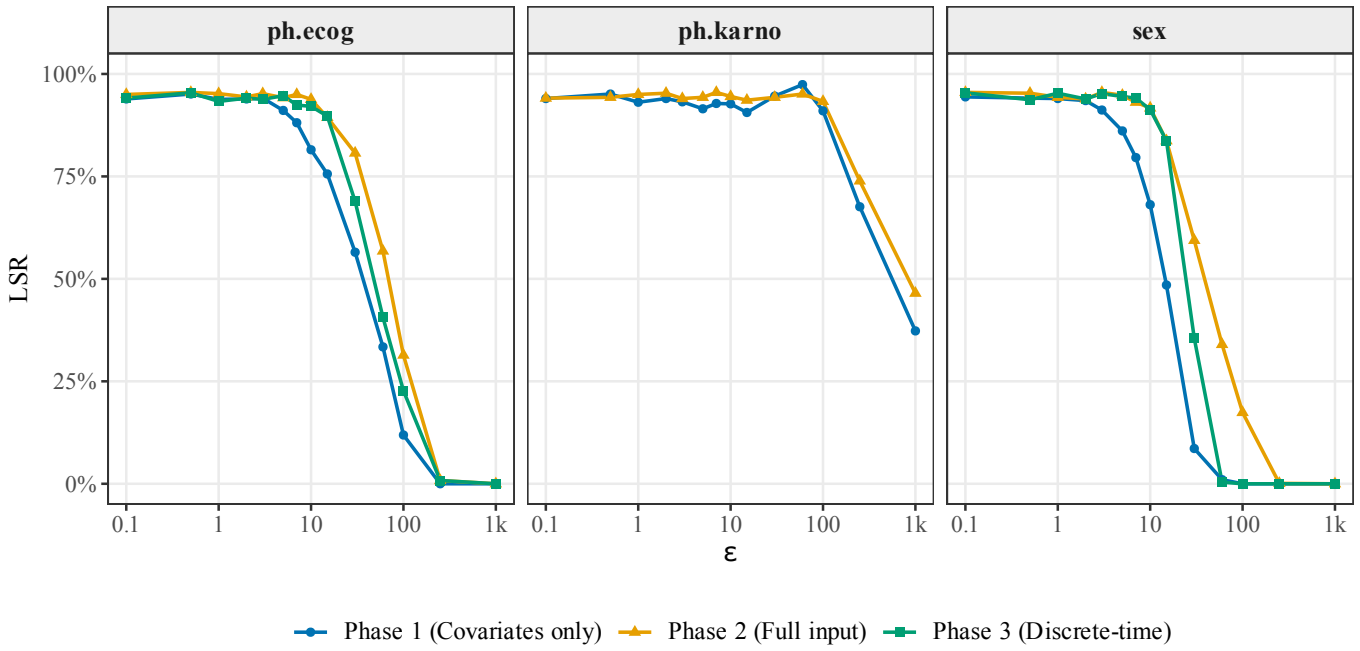


Fig. S1. Coefficient-level LSR for the lung dataset, Phases 1–3. The three coefficients significant at baseline (`sex`, `ph.ecog`, `ph.karno`) are shown separately. Recovery speed differs by coefficient according to effect size and sensitivity, which the dataset-level mean of Fig. 1 of the main manuscript conceals.

TABLE S1

LOSS OF SIGNIFICANCE RATE (LSR) FOR PHASES 1–3, ALL DATASETS × ALL ε. LSR IS THE MEAN OVER $B = 1{,}000$ ITERATIONS OF THE PROPORTION OF COEFFICIENTS SIGNIFICANT AT BASELINE ($p < 0.05$ AT $\varepsilon = \infty$) THAT BECAME NON-SIGNIFICANT AFTER PERTURBATION.

Dataset	ε	Phase 1	Phase 2	Phase 3
lung	0.1	0.941	0.949	0.948
lung	0.5	0.948	0.950	0.946
lung	1	0.936	0.948	0.943
lung	2	0.938	0.945	0.940
lung	3	0.928	0.949	0.945
lung	5	0.896	0.944	0.946
lung	7	0.868	0.945	0.932
lung	10	0.808	0.933	0.917
lung	15	0.716	0.889	0.867
lung	30	0.532	0.781	0.522
lung	60	0.439	0.620	0.205
lung	100	0.343	0.474	0.114
lung	250	0.225	0.249	0.004
lung	1000	0.124	0.155	0.000
lung	Inf	0.000	0.000	0.000
pbc	0.1	0.943	0.947	0.938
pbc	0.5	0.936	0.941	0.941
pbc	1	0.938	0.943	0.941
pbc	2	0.936	0.947	0.941
pbc	3	0.935	0.944	0.947
pbc	5	0.923	0.945	0.940
pbc	7	0.915	0.946	0.941
pbc	10	0.883	0.940	0.939
pbc	15	0.831	0.928	0.932
pbc	30	0.639	0.863	0.871
pbc	60	0.467	0.687	0.619
pbc	100	0.362	0.553	0.408
pbc	250	0.194	0.324	0.201
pbc	1000	0.013	0.059	0.049
pbc	Inf	0.000	0.000	0.000
colon	0.1	0.944	0.954	0.947
colon	0.5	0.950	0.954	0.948
colon	1	0.945	0.944	0.947
colon	2	0.931	0.947	0.951
colon	3	0.914	0.948	0.953
colon	5	0.849	0.948	0.946
colon	7	0.779	0.940	0.944
colon	10	0.686	0.917	0.920
colon	15	0.563	0.823	0.831
colon	30	0.305	0.483	0.486
colon	60	0.171	0.225	0.193
colon	100	0.155	0.165	0.164
colon	250	0.102	0.109	0.119
colon	1000	0.002	0.009	0.013
colon	Inf	0.000	0.000	0.000
rotterdam	0.1	0.947	0.955	0.953
rotterdam	0.5	0.949	0.951	0.947
rotterdam	1	0.934	0.958	0.954
rotterdam	2	0.900	0.953	0.944
rotterdam	3	0.836	0.950	0.948
rotterdam	5	0.671	0.945	0.947
rotterdam	7	0.505	0.922	0.928
rotterdam	10	0.395	0.851	0.861
rotterdam	15	0.340	0.607	0.594
rotterdam	30	0.302	0.339	0.317
rotterdam	60	0.251	0.280	0.264
rotterdam	100	0.181	0.225	0.197
rotterdam	250	0.101	0.133	0.092
rotterdam	1000	0.030	0.054	0.003
rotterdam	Inf	0.000	0.000	0.000
flchain	0.1	0.952	0.954	0.950
flchain	0.5	0.937	0.952	0.950
flchain	1	0.898	0.955	0.945
flchain	2	0.776	0.945	0.955
flchain	3	0.726	0.937	0.940
flchain	5	0.708	0.907	0.898
flchain	7	0.695	0.805	0.794
flchain	10	0.686	0.713	0.649
flchain	15	0.646	0.698	0.623
flchain	30	0.378	0.566	0.441
flchain	60	0.207	0.251	0.011
flchain	100	0.095	0.182	0.000
flchain	250	0.001	0.012	0.000
flchain	1000	0.000	0.000	0.000
flchain	Inf	0.000	0.000	0.000

TABLE S2
TEST C-INDEX, MEAN (SD), FOR PHASES 1–3 AND OUTPUT PERTURBATION WITH DFBETA SENSITIVITY, ALL DATASETS × ALL ε. 70/30 STRATIFIED SPLITS, $B = 1{,}000$ MONTE CARLO CROSS-VALIDATION ITERATIONS.

Dataset	ε	Phase 1	Phase 2	Phase 3	Output
lung	0.1	0.499 (0.098)	0.497 (0.096)	0.504 (0.097)	0.498 (0.094)
lung	0.5	0.512 (0.095)	0.499 (0.096)	0.499 (0.097)	0.530 (0.092)
lung	1	0.512 (0.095)	0.500 (0.094)	0.504 (0.096)	0.543 (0.086)
lung	2	0.526 (0.092)	0.499 (0.097)	0.503 (0.096)	0.568 (0.077)
lung	3	0.541 (0.092)	0.508 (0.097)	0.502 (0.098)	0.578 (0.071)
lung	5	0.558 (0.088)	0.518 (0.094)	0.511 (0.097)	0.597 (0.057)
lung	7	0.574 (0.078)	0.520 (0.097)	0.518 (0.094)	0.606 (0.051)
lung	10	0.594 (0.064)	0.539 (0.089)	0.538 (0.090)	0.610 (0.049)
lung	15	0.607 (0.057)	0.561 (0.081)	0.569 (0.081)	0.612 (0.046)
lung	30	0.618 (0.050)	0.598 (0.061)	0.615 (0.053)	0.613 (0.046)
lung	60	0.618 (0.047)	0.613 (0.051)	0.623 (0.047)	0.614 (0.046)
lung	100	0.618 (0.047)	0.616 (0.048)	0.623 (0.048)	0.614 (0.046)
lung	250	0.614 (0.046)	0.615 (0.047)	0.619 (0.047)	0.614 (0.046)
lung	1000	0.614 (0.046)	0.615 (0.046)	0.618 (0.047)	0.614 (0.046)
lung	Inf	0.614 (0.046)	0.614 (0.046)	0.618 (0.047)	0.614 (0.046)
pbc	0.1	0.508 (0.150)	0.496 (0.145)	0.496 (0.146)	0.512 (0.156)
pbc	0.5	0.510 (0.148)	0.507 (0.146)	0.496 (0.148)	0.544 (0.154)
pbc	1	0.532 (0.153)	0.508 (0.144)	0.496 (0.147)	0.595 (0.150)
pbc	2	0.569 (0.140)	0.503 (0.146)	0.501 (0.150)	0.669 (0.121)
pbc	3	0.594 (0.133)	0.500 (0.144)	0.512 (0.146)	0.715 (0.104)
pbc	5	0.642 (0.118)	0.514 (0.148)	0.517 (0.144)	0.778 (0.064)
pbc	7	0.690 (0.094)	0.530 (0.150)	0.528 (0.146)	0.803 (0.043)
pbc	10	0.731 (0.071)	0.558 (0.138)	0.550 (0.145)	0.816 (0.035)
pbc	15	0.763 (0.052)	0.622 (0.123)	0.605 (0.131)	0.824 (0.032)
pbc	30	0.795 (0.036)	0.750 (0.059)	0.750 (0.058)	0.829 (0.030)
pbc	60	0.814 (0.032)	0.798 (0.037)	0.805 (0.036)	0.830 (0.030)
pbc	100	0.824 (0.030)	0.814 (0.032)	0.821 (0.030)	0.831 (0.030)
pbc	250	0.832 (0.029)	0.830 (0.029)	0.833 (0.029)	0.831 (0.030)
pbc	1000	0.831 (0.030)	0.831 (0.030)	0.833 (0.029)	0.831 (0.030)
pbc	Inf	0.831 (0.030)	0.831 (0.030)	0.833 (0.029)	0.831 (0.030)
colon	0.1	0.503 (0.061)	0.499 (0.062)	0.498 (0.060)	0.512 (0.079)
colon	0.5	0.507 (0.060)	0.498 (0.062)	0.501 (0.060)	0.550 (0.074)
colon	1	0.519 (0.060)	0.502 (0.063)	0.500 (0.060)	0.588 (0.063)
colon	2	0.537 (0.057)	0.504 (0.062)	0.500 (0.063)	0.621 (0.044)
colon	3	0.555 (0.055)	0.504 (0.060)	0.503 (0.062)	0.640 (0.032)
colon	5	0.579 (0.045)	0.518 (0.060)	0.510 (0.062)	0.650 (0.024)
colon	7	0.601 (0.036)	0.530 (0.058)	0.525 (0.058)	0.655 (0.021)
colon	10	0.619 (0.029)	0.552 (0.055)	0.550 (0.055)	0.657 (0.020)
colon	15	0.637 (0.023)	0.593 (0.041)	0.589 (0.043)	0.658 (0.020)
colon	30	0.652 (0.021)	0.642 (0.023)	0.643 (0.022)	0.659 (0.020)
colon	60	0.657 (0.020)	0.655 (0.020)	0.656 (0.020)	0.659 (0.020)
colon	100	0.658 (0.020)	0.656 (0.020)	0.658 (0.020)	0.659 (0.020)
colon	250	0.658 (0.020)	0.658 (0.020)	0.658 (0.020)	0.659 (0.020)
colon	1000	0.659 (0.020)	0.659 (0.020)	0.659 (0.020)	0.659 (0.020)
colon	Inf	0.659 (0.020)	0.659 (0.020)	0.659 (0.020)	0.659 (0.020)
rotterdam	0.1	0.507 (0.061)	0.502 (0.059)	0.498 (0.059)	0.516 (0.085)
rotterdam	0.5	0.514 (0.060)	0.503 (0.060)	0.495 (0.060)	0.582 (0.075)
rotterdam	1	0.533 (0.055)	0.502 (0.061)	0.499 (0.059)	0.624 (0.059)
rotterdam	2	0.558 (0.050)	0.509 (0.061)	0.502 (0.063)	0.656 (0.031)
rotterdam	3	0.579 (0.041)	0.510 (0.060)	0.509 (0.060)	0.667 (0.017)
rotterdam	5	0.606 (0.025)	0.526 (0.058)	0.526 (0.057)	0.672 (0.013)
rotterdam	7	0.616 (0.019)	0.546 (0.054)	0.545 (0.055)	0.673 (0.012)
rotterdam	10	0.625 (0.015)	0.579 (0.042)	0.578 (0.040)	0.674 (0.011)
rotterdam	15	0.633 (0.013)	0.613 (0.022)	0.615 (0.021)	0.675 (0.011)
rotterdam	30	0.642 (0.012)	0.636 (0.013)	0.637 (0.013)	0.675 (0.011)
rotterdam	60	0.655 (0.012)	0.648 (0.013)	0.651 (0.012)	0.675 (0.011)
rotterdam	100	0.665 (0.012)	0.660 (0.012)	0.662 (0.012)	0.675 (0.011)
rotterdam	250	0.673 (0.011)	0.672 (0.012)	0.674 (0.011)	0.675 (0.011)
rotterdam	1000	0.675 (0.011)	0.675 (0.011)	0.676 (0.011)	0.675 (0.011)
rotterdam	Inf	0.675 (0.011)	0.675 (0.011)	0.677 (0.011)	0.675 (0.011)
flchain	0.1	0.520 (0.146)	0.498 (0.146)	0.505 (0.143)	0.635 (0.131)
flchain	0.5	0.573 (0.132)	0.503 (0.144)	0.507 (0.142)	0.760 (0.045)
flchain	1	0.628 (0.115)	0.513 (0.145)	0.512 (0.142)	0.780 (0.021)
flchain	2	0.700 (0.066)	0.535 (0.144)	0.520 (0.143)	0.788 (0.011)
flchain	3	0.734 (0.047)	0.552 (0.140)	0.552 (0.143)	0.789 (0.009)
flchain	5	0.760 (0.026)	0.628 (0.115)	0.615 (0.121)	0.790 (0.008)
flchain	7	0.771 (0.016)	0.692 (0.072)	0.678 (0.087)	0.790 (0.008)
flchain	10	0.777 (0.011)	0.745 (0.038)	0.740 (0.040)	0.790 (0.008)
flchain	15	0.780 (0.009)	0.773 (0.014)	0.772 (0.015)	0.790 (0.008)
flchain	30	0.783 (0.008)	0.782 (0.008)	0.782 (0.008)	0.791 (0.008)
flchain	60	0.786 (0.008)	0.785 (0.008)	0.785 (0.008)	0.791 (0.008)
flchain	100	0.788 (0.008)	0.787 (0.008)	0.787 (0.008)	0.791 (0.008)
flchain	250	0.790 (0.008)	0.790 (0.008)	0.790 (0.008)	0.791 (0.008)
flchain	1000	0.790 (0.008)	0.790 (0.008)	0.791 (0.008)	0.791 (0.008)
flchain	Inf	0.791 (0.008)	0.791 (0.008)	0.791 (0.008)	0.791 (0.008)

TABLE S3
MEAN FALSE-POSITIVE RATE BY ε AND PHASE (INPUT PERTURBATION ONLY), AT ALL 15 BUDGETS; THE RIGHT-TAIL THRESHOLDS $\varepsilon^*_{\mathrm{FPR}\le 0.10}$ OF TABLE V OF THE MAIN MANUSCRIPT CAN BE READ FROM THESE ROWS.

Dataset	ε	Phase 1	Phase 2	Phase 3
lung	0.1	0.065	0.050	0.053
lung	0.5	0.065	0.052	0.054
lung	1	0.053	0.051	0.051
lung	2	0.054	0.057	0.060
lung	3	0.066	0.050	0.057
lung	5	0.064	0.046	0.054
lung	7	0.074	0.057	0.062
lung	10	0.071	0.053	0.058
lung	15	0.078	0.064	0.060
lung	30	0.092	0.068	0.077
lung	60	0.093	0.078	0.095
lung	100	0.090	0.079	0.078
lung	250	0.068	0.074	0.042
lung	1000	0.013	0.028	0.001
lung	∞	0.000	0.000	0.000
pbc	0.1	0.055	0.057	0.057
pbc	0.5	0.058	0.052	0.064
pbc	1	0.062	0.052	0.057
pbc	2	0.062	0.057	0.062
pbc	3	0.070	0.055	0.058
pbc	5	0.081	0.058	0.055
pbc	7	0.094	0.061	0.059
pbc	10	0.137	0.060	0.060
pbc	15	0.208	0.071	0.072
pbc	30	0.337	0.158	0.155
pbc	60	0.339	0.260	0.309
pbc	100	0.218	0.250	0.284
pbc	250	0.048	0.115	0.078
pbc	1000	0.000	0.002	0.000
pbc	∞	0.000	0.000	0.000
colon	0.1	0.056	0.050	0.048
colon	0.5	0.049	0.052	0.054
colon	1	0.050	0.051	0.050
colon	2	0.051	0.049	0.055
colon	3	0.050	0.046	0.051
colon	5	0.060	0.051	0.051
colon	7	0.060	0.053	0.053
colon	10	0.069	0.051	0.051
colon	15	0.100	0.055	0.054
colon	30	0.174	0.094	0.100
colon	60	0.155	0.064	0.133
colon	100	0.141	0.023	0.115
colon	250	0.109	0.027	0.057
colon	1000	0.031	0.031	0.000
colon	∞	0.000	0.000	0.000
rotterdam	0.1	0.051	0.041	0.048
rotterdam	0.5	0.045	0.046	0.050
rotterdam	1	0.052	0.045	0.054
rotterdam	2	0.050	0.049	0.050
rotterdam	3	0.058	0.058	0.049
rotterdam	5	0.060	0.051	0.049
rotterdam	7	0.074	0.057	0.059
rotterdam	10	0.109	0.055	0.057
rotterdam	15	0.145	0.076	0.072
rotterdam	30	0.120	0.128	0.122
rotterdam	60	0.025	0.056	0.029
rotterdam	100	0.027	0.023	0.031
rotterdam	250	0.016	0.015	0.024
rotterdam	1000	0.000	0.000	0.000
rotterdam	∞	0.000	0.000	0.000
flchain	0.1	0.055	0.051	0.045
flchain	0.5	0.051	0.041	0.036
flchain	1	0.056	0.051	0.058
flchain	2	0.059	0.049	0.048
flchain	3	0.064	0.044	0.044
flchain	5	0.082	0.050	0.044
flchain	7	0.100	0.054	0.054
flchain	10	0.148	0.069	0.057
flchain	15	0.238	0.103	0.081
flchain	30	0.514	0.333	0.213
flchain	60	0.612	0.610	0.483
flchain	100	0.619	0.607	0.427
flchain	250	0.359	0.432	0.202
flchain	1000	0.291	0.310	0.000
flchain	∞	0.000	0.000	0.000

S3. FALSE POSITIVES: ROBUSTNESS AND MECHANISM

These are the analyses behind Section IV-D of the main manuscript. Fig. S2 and Table S4 establish that the inverted-U pattern is not an artifact of the significance level; Fig. S3 decomposes its mechanism; Fig. S4 isolates the effect of cohort size by subsampling.

Definitions behind Fig. S3. For one coefficient at one budget, with $\hat{\beta}^{(b)}$ and $\mathrm{SE}^{(b)}$ the estimate and its standard error in iteration b, the center and dispersion of the perturbed Wald statistic are $z_{\mathrm{eff}} = \overline{\hat{\beta}}/\overline{\mathrm{SE}}$ and $d_{\mathrm{eff}} = \mathrm{sd}(\hat{\beta})/\overline{\mathrm{SE}}$, the bar denoting the mean over iterations. Approximating the perturbed statistic by $\mathcal{N}(z_{\mathrm{eff}}, d^2_{\mathrm{eff}})$ reproduces the observed FPR as $\Pr(|\mathcal{N}(z_{\mathrm{eff}}, d^2_{\mathrm{eff}})| > 1.96)$, and $z_{\mathrm{eff}} = (z_{\mathrm{base}} + s)/r_{\mathrm{SE}}$, where s is the shift of the mean coefficient in units of the baseline standard error and r_{SE} the ratio of perturbed to baseline standard error. The correlations quoted in Section IV-D are Spearman rank correlations of peak FPR with $|z_{\mathrm{eff}}|$ and d_{eff} at each coefficient's peak budget, over the 27 Phase 1 coefficients that form the FPR denominators at $\alpha = 0.05$ (per-coefficient values of z_{eff}, d_{eff}, and the peak in `fpr_mechanism_peak_decomp.csv`, repository). The FPR is, by this construction, a function of z_{eff} and d_{eff}; therefore, the correlation with $|z_{\mathrm{eff}}|$ is partly expected. The decomposition into shift and standard-error components is the informative part. Replacing only r_{SE} or only s by its perturbed value at that budget reproduces at least half of the observed peak for 4 and 10 coefficients, respectively; for 3, either suffices, and for 10 both are needed (panel B of Fig. S3); the shift of the center is therefore the more often sufficient component (13 against 7 coefficients, the counts of Section IV-D of the main manuscript, the 3 being counted in both).

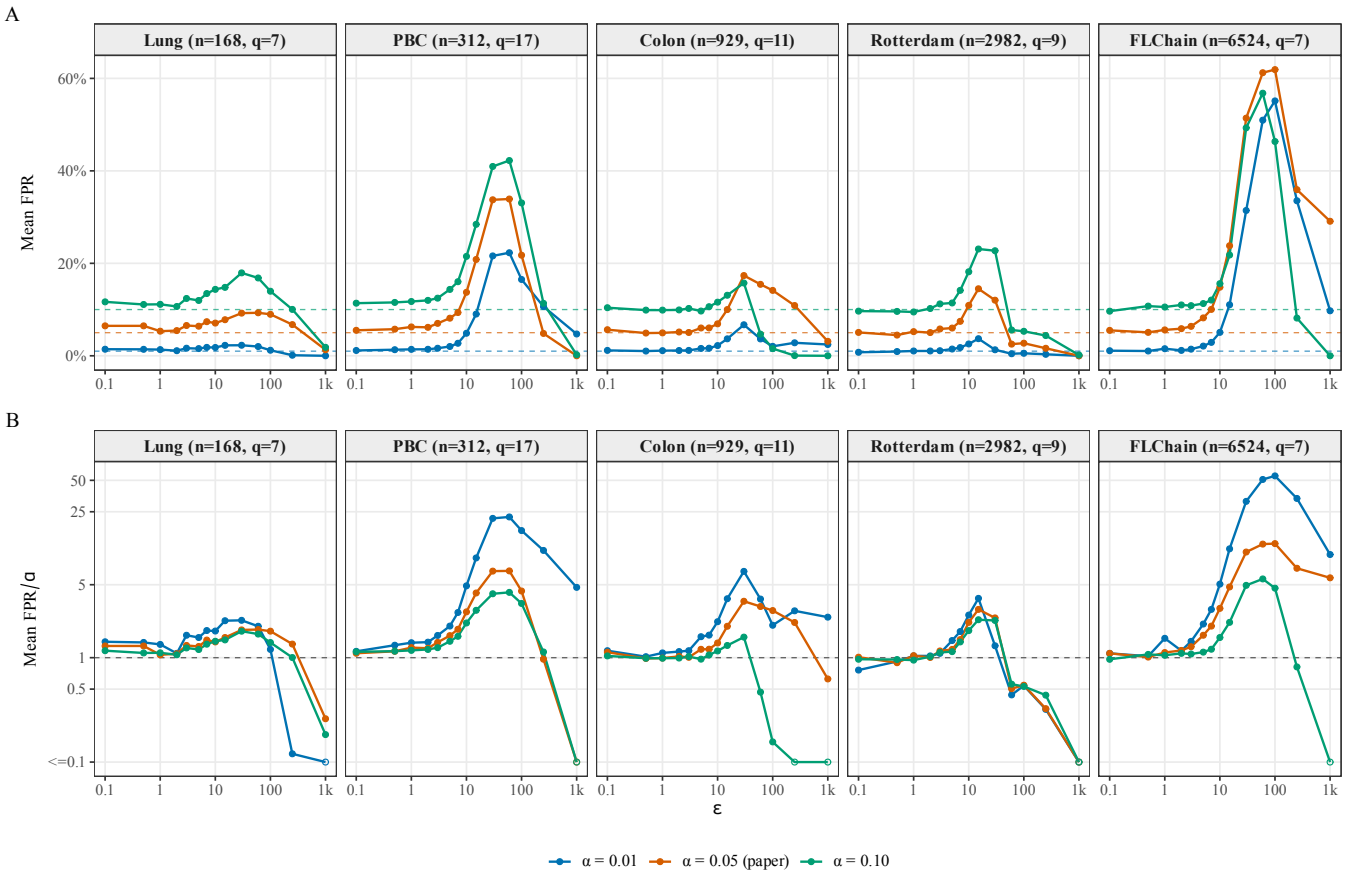


Fig. S2. Mean FPR against ε at three significance levels ($\alpha \in \{0.01, 0.05, 0.10\}$). Panel B plots the ratio of the FPR to α on a common logarithmic axis, so that the three levels are directly comparable. The peak exceeds α in all 45 dataset-by-phase-by-α conditions.

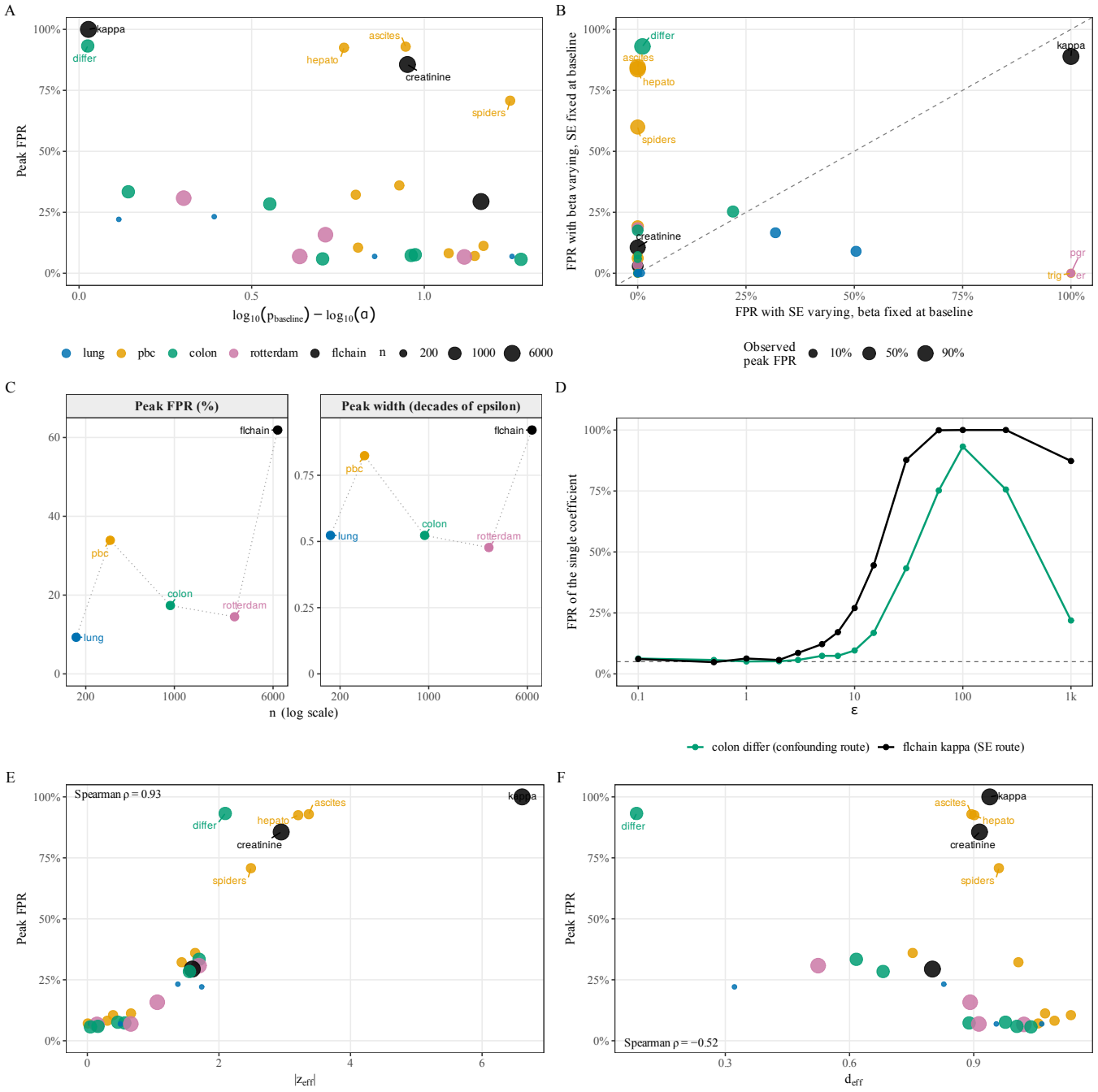


Fig. S3. Mechanism of the FPR peak, Phase 1 at $\alpha = 0.05$. (A) Peak FPR of each of the 27 coefficients against its distance from the significance threshold, $\log_{10} p_{\text{baseline}} - \log_{10} \alpha$; point size is n. (B) The two routes to a false positive at each coefficient's peak budget: the FPR obtained when only the standard error varies, and $\hat{\beta}$ is held at its baseline value (horizontal) against the FPR when only $\hat{\beta}$ varies (vertical); the sufficiency counts of Section IV-D (13, 7, and 10) are read from this panel. (C) Dataset-level peak height and width against n. (D) colon `differ` and flchain `kappa`, which have almost the same baseline p-value and peak at the same budget by different routes. (E, F) Peak FPR of the same 27 coefficients against the center $|z_{\text{eff}}|$ and the dispersion d_{eff} of the perturbed Wald statistic at each coefficient's peak budget, with the Spearman rank correlations quoted in Section IV-D (0.93 and -0.52); colors and point sizes as in (A). The per-coefficient values are in `fpr_mechanism_peak_decomp.csv` in this repository.

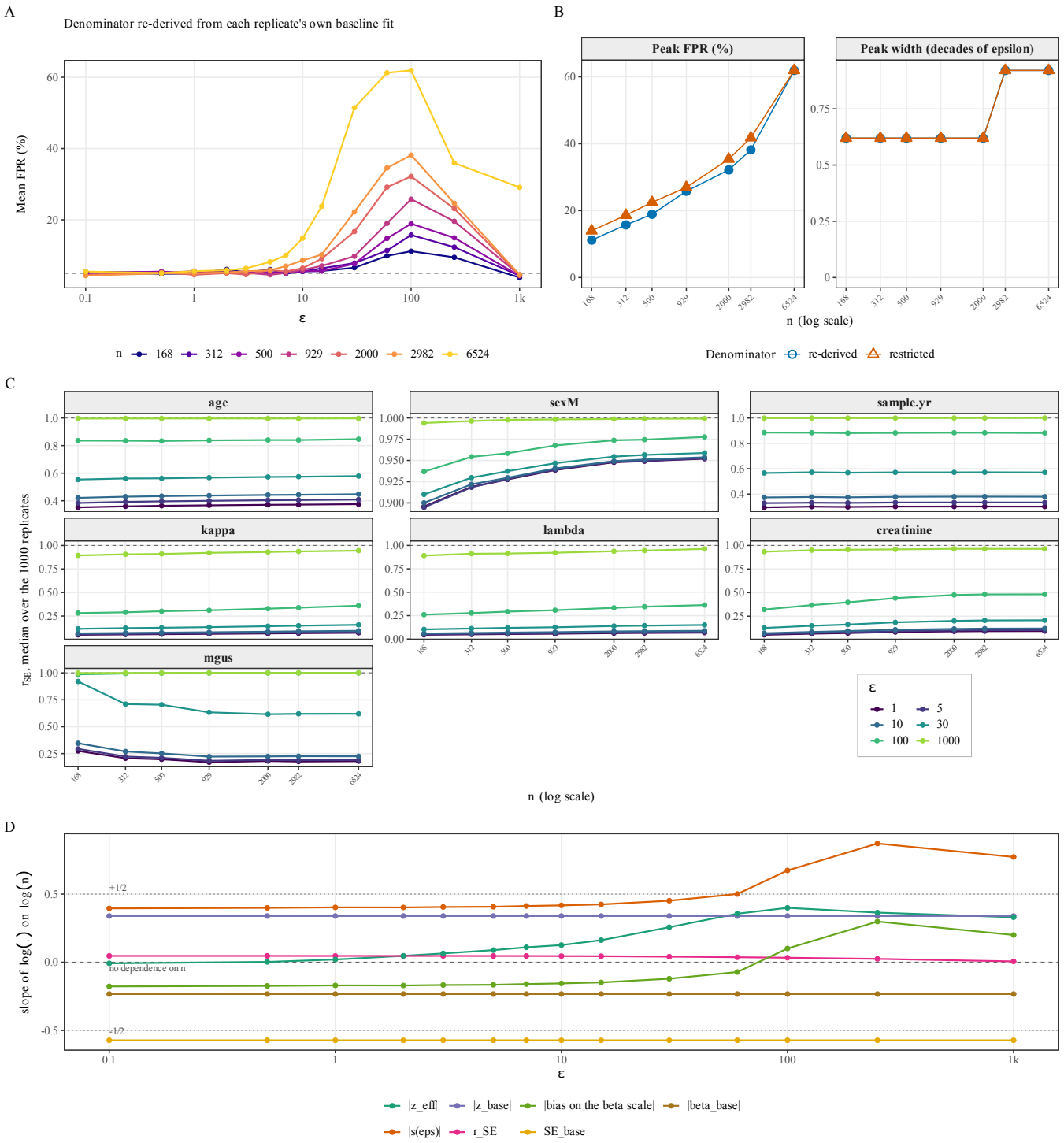


Fig. S4. FPR against cohort size, obtained by subsampling flchain into seven sizes from $n = 168$ to 6,524 with the covariate set and the noise scale held fixed, so that only n varies (112,000 fits: the 15 budgets and one unperturbed baseline fit for each of 7 sizes $\times$ 1,000 subsamples). The peak rises monotonically from 0.112 to 0.619 (rank correlation 1.000). The driver is the *pre-perturbation* standard error, not a stronger shrinkage of standard errors in larger samples: the shrinkage ratio is nearly independent of n and its sign is opposite to that hypothesis.

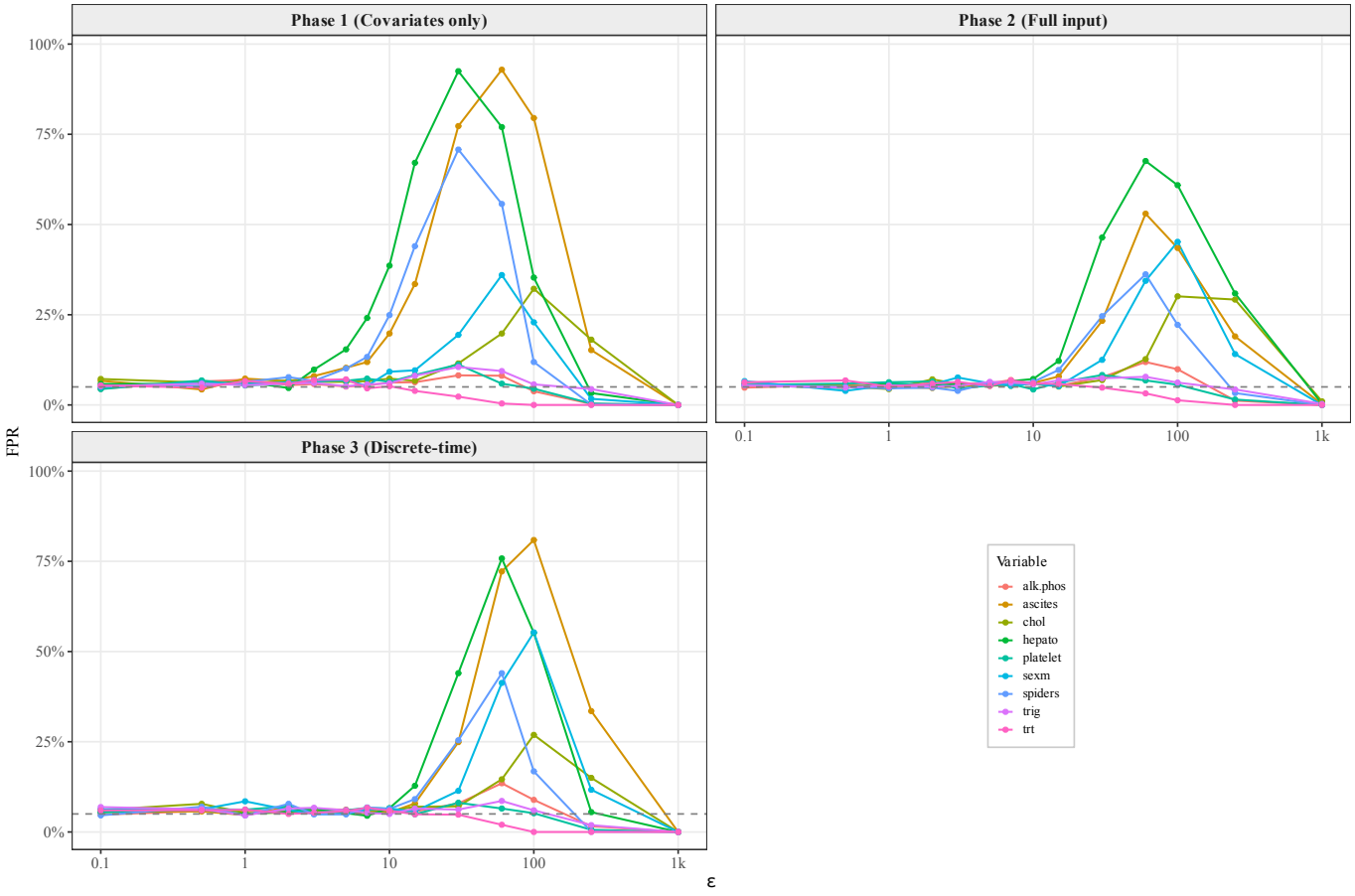


Fig. S5. Coefficient-level FPR for the pbc dataset ($q = 17$) by Phase, with a common y-axis; the dashed line is $\alpha = 0.05$. The peak in the intermediate range is carried by coefficients whose baseline p-values lie near the threshold, and every curve returns to 0 as $\varepsilon \to \infty$. This is the coefficient-level view of Fig. 3 of the main manuscript.

TABLE S4: Significance-level sensitivity: LSR and FPR recomputed at $\alpha \in \{0.01, 0.05, 0.10\}$ for every dataset, phase, and ε (225 rows). Since the denominator set is redrawn at each α, the mean FPR can fall when α is relaxed (33 cells); this reflects a change in which coefficients are averaged. With the denominator held at the $\alpha = 0.05$ set, the median inflation of the peak over α is 4.53, 2.90 and 2.92 at $\alpha = 0.01$, 0.05 and 0.10 (6.70, 2.90 and 2.06 with per-α denominators).

Dataset	Phase	ε	FPR (α=0.01)	FPR (α=0.05)	FPR (α=0.10)	LSR (α=0.01)	LSR (α=0.05)	LSR (α=0.10)
lung	Phase 1	0.1	0.0142	0.0648	0.1167	0.9910	0.9410	0.8875
lung	Phase 1	0.5	0.0140	0.0648	0.1110	0.9905	0.9477	0.8833
lung	Phase 1	1	0.0134	0.0533	0.1113	0.9895	0.9363	0.8860
lung	Phase 1	2	0.0110	0.0545	0.1067	0.9835	0.9380	0.8858
lung	Phase 1	3	0.0164	0.0655	0.1240	0.9830	0.9283	0.8790
lung	Phase 1	5	0.0156	0.0640	0.1197	0.9645	0.8957	0.8425
lung	Phase 1	7	0.0182	0.0738	0.1347	0.9475	0.8683	0.8120
lung	Phase 1	10	0.0180	0.0708	0.1437	0.9195	0.8077	0.7558
lung	Phase 1	15	0.0226	0.0780	0.1480	0.8645	0.7157	0.6740
lung	Phase 1	30	0.0228	0.0923	0.1793	0.7270	0.5323	0.5605
lung	Phase 1	60	0.0200	0.0930	0.1683	0.5890	0.4393	0.4963
lung	Phase 1	100	0.0120	0.0898	0.1397	0.4095	0.3430	0.3898
lung	Phase 1	250	0.0012	0.0675	0.1003	0.0825	0.2253	0.1703
lung	Phase 1	1000	0.0000	0.0130	0.0183	0.0000	0.1243	0.0030
lung	Phase 1	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
lung	Phase 2	0.1	0.0088	0.0500	0.1013	0.9895	0.9487	0.8938
lung	Phase 2	0.5	0.0116	0.0525	0.0997	0.9910	0.9503	0.8913
lung	Phase 2	1	0.0104	0.0508	0.1073	0.9890	0.9483	0.9000
lung	Phase 2	2	0.0128	0.0568	0.1103	0.9930	0.9453	0.8930
lung	Phase 2	3	0.0100	0.0498	0.1047	0.9955	0.9490	0.8963
lung	Phase 2	5	0.0100	0.0465	0.0997	0.9900	0.9443	0.8983
lung	Phase 2	7	0.0114	0.0573	0.1113	0.9895	0.9453	0.8890
lung	Phase 2	10	0.0098	0.0530	0.1033	0.9820	0.9333	0.8728
lung	Phase 2	15	0.0140	0.0635	0.1163	0.9650	0.8890	0.8365
lung	Phase 2	30	0.0148	0.0678	0.1367	0.8845	0.7813	0.7255
lung	Phase 2	60	0.0166	0.0778	0.1540	0.7590	0.6197	0.6030

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Table S4 (continued)

Dataset	Phase	ε	FPR (α=0.01)	FPR (α=0.05)	FPR (α=0.10)	LSR (α=0.01)	LSR (α=0.05)	LSR (α=0.10)
lung	Phase 2	100	0.0156	0.0785	0.1413	0.6285	0.4737	0.4905
lung	Phase 2	250	0.0040	0.0738	0.1113	0.2405	0.2493	0.2528
lung	Phase 2	1000	0.0000	0.0275	0.0427	0.0025	0.1550	0.0185
lung	Phase 2	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
lung	Phase 3	0.1	0.0112	0.0535	0.1100	0.9875	0.9475	0.8943
lung	Phase 3	0.5	0.0118	0.0538	0.1123	0.9915	0.9455	0.9033
lung	Phase 3	1	0.0088	0.0510	0.1043	0.9895	0.9430	0.8980
lung	Phase 3	2	0.0120	0.0598	0.1107	0.9850	0.9400	0.8817
lung	Phase 3	3	0.0116	0.0565	0.1133	0.9885	0.9450	0.8917
lung	Phase 3	5	0.0128	0.0545	0.1103	0.9875	0.9460	0.8923
lung	Phase 3	7	0.0100	0.0620	0.1147	0.9855	0.9325	0.8777
lung	Phase 3	10	0.0118	0.0583	0.1137	0.9810	0.9170	0.8687
lung	Phase 3	15	0.0164	0.0600	0.1187	0.9550	0.8670	0.8243
lung	Phase 3	30	0.0230	0.0773	0.1657	0.7825	0.5220	0.5593
lung	Phase 3	60	0.0212	0.0950	0.1783	0.4545	0.2050	0.3790
lung	Phase 3	100	0.0090	0.0778	0.1410	0.2990	0.1135	0.3050
lung	Phase 3	250	0.0002	0.0415	0.0873	0.0905	0.0040	0.2007
lung	Phase 3	1000	0.0000	0.0008	0.0030	0.0000	0.0000	0.1293
lung	Phase 3	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
pbc	Phase 1	0.1	0.0115	0.0551	0.1138	0.9876	0.9431	0.8883
pbc	Phase 1	0.5	0.0132	0.0576	0.1154	0.9864	0.9364	0.8820
pbc	Phase 1	1	0.0139	0.0624	0.1174	0.9880	0.9381	0.8810
pbc	Phase 1	2	0.0141	0.0616	0.1199	0.9836	0.9361	0.8801
pbc	Phase 1	3	0.0163	0.0701	0.1247	0.9836	0.9354	0.8809
pbc	Phase 1	5	0.0201	0.0813	0.1436	0.9806	0.9228	0.8643
pbc	Phase 1	7	0.0271	0.0936	0.1602	0.9786	0.9150	0.8551
pbc	Phase 1	10	0.0488	0.1373	0.2150	0.9642	0.8831	0.8119
pbc	Phase 1	15	0.0903	0.2083	0.2843	0.9372	0.8311	0.7410
pbc	Phase 1	30	0.2160	0.3374	0.4095	0.7970	0.6385	0.5349
pbc	Phase 1	60	0.2229	0.3391	0.4223	0.6232	0.4673	0.3748
pbc	Phase 1	100	0.1651	0.2176	0.3307	0.5356	0.3616	0.2779
pbc	Phase 1	250	0.1065	0.0483	0.1134	0.4118	0.1944	0.1126
pbc	Phase 1	1000	0.0472	0.0000	0.0022	0.2378	0.0134	0.0004
pbc	Phase 1	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
pbc	Phase 2	0.1	0.0103	0.0571	0.1091	0.9894	0.9465	0.8938
pbc	Phase 2	0.5	0.0118	0.0523	0.1050	0.9870	0.9408	0.8919
pbc	Phase 2	1	0.0108	0.0524	0.1053	0.9890	0.9425	0.8853
pbc	Phase 2	2	0.0113	0.0571	0.1073	0.9908	0.9470	0.8956
pbc	Phase 2	3	0.0121	0.0553	0.1104	0.9876	0.9439	0.8915
pbc	Phase 2	5	0.0138	0.0580	0.1148	0.9868	0.9451	0.8911
pbc	Phase 2	7	0.0123	0.0607	0.1130	0.9860	0.9460	0.8883
pbc	Phase 2	10	0.0133	0.0596	0.1124	0.9860	0.9404	0.8860
pbc	Phase 2	15	0.0198	0.0713	0.1293	0.9806	0.9284	0.8734
pbc	Phase 2	30	0.0572	0.1577	0.2336	0.9522	0.8628	0.7876
pbc	Phase 2	60	0.1293	0.2596	0.3516	0.8520	0.6870	0.5819
pbc	Phase 2	100	0.1322	0.2499	0.3452	0.7616	0.5534	0.4468
pbc	Phase 2	250	0.1049	0.1150	0.2021	0.5734	0.3241	0.2161
pbc	Phase 2	1000	0.0554	0.0020	0.0206	0.3546	0.0593	0.0130
pbc	Phase 2	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
pbc	Phase 3	0.1	0.0145	0.0570	0.1144	0.9865	0.9379	0.8870
pbc	Phase 3	0.5	0.0133	0.0637	0.1180	0.9855	0.9411	0.8864
pbc	Phase 3	1	0.0133	0.0567	0.1097	0.9853	0.9409	0.8879
pbc	Phase 3	2	0.0143	0.0623	0.1134	0.9850	0.9410	0.8864
pbc	Phase 3	3	0.0142	0.0578	0.1119	0.9858	0.9466	0.8924
pbc	Phase 3	5	0.0132	0.0553	0.1087	0.9853	0.9398	0.8815
pbc	Phase 3	7	0.0153	0.0589	0.1132	0.9890	0.9411	0.8880
pbc	Phase 3	10	0.0134	0.0601	0.1190	0.9828	0.9385	0.8799
pbc	Phase 3	15	0.0183	0.0717	0.1352	0.9838	0.9319	0.8770
pbc	Phase 3	30	0.0568	0.1553	0.2332	0.9653	0.8713	0.7951
pbc	Phase 3	60	0.1701	0.3094	0.4006	0.8548	0.6190	0.5161
pbc	Phase 3	100	0.1753	0.2836	0.3934	0.6970	0.4083	0.3171
pbc	Phase 3	250	0.1153	0.0777	0.1792	0.4793	0.2014	0.1149
pbc	Phase 3	1000	0.0608	0.0000	0.0013	0.2318	0.0493	0.0014
pbc	Phase 3	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
colon	Phase 1	0.1	0.0117	0.0564	0.1040	0.9897	0.9444	0.8886
colon	Phase 1	0.5	0.0102	0.0491	0.0988	0.9903	0.9498	0.8970
colon	Phase 1	1	0.0111	0.0496	0.0988	0.9877	0.9446	0.8917
colon	Phase 1	2	0.0114	0.0514	0.0992	0.9793	0.9312	0.8821
colon	Phase 1	3	0.0117	0.0504	0.1024	0.9683	0.9138	0.8684
colon	Phase 1	5	0.0159	0.0601	0.0968	0.9213	0.8494	0.8007
colon	Phase 1	7	0.0164	0.0603	0.1062	0.8363	0.7792	0.7466
colon	Phase 1	10	0.0221	0.0691	0.1160	0.6947	0.6856	0.6747
colon	Phase 1	15	0.0367	0.1000	0.1309	0.5372	0.5630	0.5543
colon	Phase 1	30	0.0671	0.1736	0.1574	0.1990	0.3054	0.2954
colon	Phase 1	60	0.0364	0.1546	0.0468	0.0000	0.1714	0.1409
colon	Phase 1	100	0.0204	0.1414	0.0156	0.0000	0.1548	0.1000
colon	Phase 1	250	0.0281	0.1087	0.0004	0.0000	0.1016	0.0383
colon	Phase 1	1000	0.0244	0.0313	0.0000	0.0000	0.0024	0.0000
colon	Phase 1	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
colon	Phase 2	0.1	0.0106	0.0499	0.1022	0.9933	0.9538	0.9009
colon	Phase 2	0.5	0.0102	0.0521	0.1026	0.9910	0.9536	0.8964
colon	Phase 2	1	0.0100	0.0509	0.1038	0.9863	0.9438	0.8947

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Table S4 (continued)

Dataset	Phase	ε	FPR (α=0.01)	FPR (α=0.05)	FPR (α=0.10)	LSR (α=0.01)	LSR (α=0.05)	LSR (α=0.10)
colon	Phase 2	2	0.0104	0.0494	0.0994	0.9867	0.9470	0.8981
colon	Phase 2	3	0.0107	0.0460	0.0956	0.9900	0.9484	0.8983
colon	Phase 2	5	0.0103	0.0510	0.1016	0.9903	0.9480	0.9021
colon	Phase 2	7	0.0098	0.0526	0.1060	0.9860	0.9398	0.8864
colon	Phase 2	10	0.0122	0.0509	0.1046	0.9703	0.9168	0.8657
colon	Phase 2	15	0.0143	0.0551	0.0992	0.8727	0.8234	0.7821
colon	Phase 2	30	0.0353	0.0938	0.1415	0.4444	0.4831	0.5161
colon	Phase 2	60	0.0389	0.0640	0.1140	0.0190	0.2252	0.3380
colon	Phase 2	100	0.0476	0.0229	0.0648	0.0000	0.1646	0.3131
colon	Phase 2	250	0.0656	0.0270	0.0150	0.0000	0.1086	0.1793
colon	Phase 2	1000	0.0670	0.0311	0.0000	0.0000	0.0094	0.0049
colon	Phase 2	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
colon	Phase 3	0.1	0.0091	0.0476	0.0984	0.9907	0.9468	0.8949
colon	Phase 3	0.5	0.0094	0.0536	0.1044	0.9877	0.9482	0.9017
colon	Phase 3	1	0.0110	0.0501	0.1048	0.9883	0.9466	0.9014
colon	Phase 3	2	0.0121	0.0546	0.1054	0.9903	0.9514	0.8976
colon	Phase 3	3	0.0106	0.0507	0.0966	0.9887	0.9526	0.9014
colon	Phase 3	5	0.0106	0.0509	0.1018	0.9890	0.9462	0.8996
colon	Phase 3	7	0.0114	0.0530	0.0990	0.9860	0.9440	0.8944
colon	Phase 3	10	0.0094	0.0509	0.1058	0.9727	0.9200	0.8741
colon	Phase 3	15	0.0111	0.0541	0.1040	0.9013	0.8308	0.7864
colon	Phase 3	30	0.0259	0.0997	0.1400	0.4453	0.4862	0.4941
colon	Phase 3	60	0.0193	0.1326	0.0888	0.0063	0.1932	0.2043
colon	Phase 3	100	0.0070	0.1147	0.0200	0.0000	0.1636	0.1106
colon	Phase 3	250	0.0116	0.0567	0.0012	0.0000	0.1194	0.0553
colon	Phase 3	1000	0.0148	0.0000	0.0000	0.0000	0.0134	0.0001
colon	Phase 3	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
rotterdam	Phase 1	0.1	0.0076	0.0506	0.0966	0.9868	0.9471	0.8981
rotterdam	Phase 1	0.5	0.0092	0.0448	0.0960	0.9892	0.9490	0.8983
rotterdam	Phase 1	1	0.0104	0.0523	0.0948	0.9828	0.9345	0.8737
rotterdam	Phase 1	2	0.0104	0.0503	0.1023	0.9690	0.9000	0.8322
rotterdam	Phase 1	3	0.0110	0.0580	0.1123	0.9350	0.8360	0.7493
rotterdam	Phase 1	5	0.0146	0.0600	0.1143	0.7938	0.6710	0.5732
rotterdam	Phase 1	7	0.0178	0.0745	0.1415	0.5788	0.5048	0.4298
rotterdam	Phase 1	10	0.0256	0.1090	0.1818	0.3640	0.3953	0.3543
rotterdam	Phase 1	15	0.0370	0.1450	0.2310	0.2670	0.3397	0.3135
rotterdam	Phase 1	30	0.0130	0.1203	0.2273	0.1894	0.3017	0.2820
rotterdam	Phase 1	60	0.0044	0.0253	0.0558	0.1526	0.2512	0.2160
rotterdam	Phase 1	100	0.0054	0.0273	0.0530	0.0898	0.1813	0.1417
rotterdam	Phase 1	250	0.0032	0.0163	0.0438	0.0024	0.1010	0.0400
rotterdam	Phase 1	1000	0.0000	0.0000	0.0018	0.0000	0.0305	0.0000
rotterdam	Phase 1	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
rotterdam	Phase 2	0.1	0.0078	0.0415	0.0935	0.9908	0.9552	0.9092
rotterdam	Phase 2	0.5	0.0080	0.0460	0.0955	0.9898	0.9510	0.9023
rotterdam	Phase 2	1	0.0090	0.0450	0.0960	0.9918	0.9582	0.9125
rotterdam	Phase 2	2	0.0104	0.0495	0.1005	0.9904	0.9528	0.9078
rotterdam	Phase 2	3	0.0100	0.0578	0.1023	0.9892	0.9498	0.9020
rotterdam	Phase 2	5	0.0094	0.0510	0.0993	0.9882	0.9448	0.8967
rotterdam	Phase 2	7	0.0138	0.0573	0.1108	0.9770	0.9222	0.8620
rotterdam	Phase 2	10	0.0110	0.0550	0.1083	0.9396	0.8508	0.7702
rotterdam	Phase 2	15	0.0154	0.0760	0.1380	0.7102	0.6072	0.5133
rotterdam	Phase 2	30	0.0288	0.1278	0.2063	0.2600	0.3385	0.3122
rotterdam	Phase 2	60	0.0050	0.0560	0.1390	0.1748	0.2805	0.2550
rotterdam	Phase 2	100	0.0038	0.0228	0.0440	0.1336	0.2245	0.1930
rotterdam	Phase 2	250	0.0034	0.0153	0.0405	0.0206	0.1325	0.0878
rotterdam	Phase 2	1000	0.0000	0.0000	0.0058	0.0000	0.0538	0.0002
rotterdam	Phase 2	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
rotterdam	Phase 3	0.1	0.0112	0.0478	0.1040	0.9892	0.9533	0.9027
rotterdam	Phase 3	0.5	0.0088	0.0498	0.1040	0.9886	0.9470	0.8958
rotterdam	Phase 3	1	0.0106	0.0543	0.1070	0.9916	0.9538	0.9037
rotterdam	Phase 3	2	0.0088	0.0495	0.1070	0.9904	0.9443	0.8930
rotterdam	Phase 3	3	0.0082	0.0490	0.1025	0.9908	0.9485	0.8988
rotterdam	Phase 3	5	0.0118	0.0493	0.0920	0.9866	0.9470	0.8977
rotterdam	Phase 3	7	0.0110	0.0588	0.1080	0.9812	0.9278	0.8708
rotterdam	Phase 3	10	0.0100	0.0573	0.1040	0.9468	0.8607	0.7758
rotterdam	Phase 3	15	0.0140	0.0715	0.1015	0.7058	0.5942	0.5090
rotterdam	Phase 3	30	0.0280	0.1223	0.1040	0.2264	0.3167	0.2977
rotterdam	Phase 3	60	0.0042	0.0293	0.1130	0.1626	0.2640	0.2368
rotterdam	Phase 3	100	0.0064	0.0313	0.1130	0.0954	0.1972	0.1595
rotterdam	Phase 3	250	0.0086	0.0243	0.1105	0.0020	0.0923	0.0397
rotterdam	Phase 3	1000	0.0002	0.0003	0.0180	0.0000	0.0028	0.0000
rotterdam	Phase 3	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
flchain	Phase 1	0.1	0.0110	0.0550	0.0965	0.9900	0.9523	0.9020
flchain	Phase 1	0.5	0.0103	0.0507	0.1075	0.9885	0.9373	0.8852
flchain	Phase 1	1	0.0153	0.0560	0.1055	0.9675	0.8985	0.8468
flchain	Phase 1	2	0.0117	0.0587	0.1100	0.8693	0.7765	0.7504
flchain	Phase 1	3	0.0143	0.0637	0.1085	0.7715	0.7263	0.7140
flchain	Phase 1	5	0.0210	0.0820	0.1130	0.7423	0.7075	0.6902
flchain	Phase 1	7	0.0290	0.1003	0.1205	0.7385	0.6953	0.6678
flchain	Phase 1	10	0.0507	0.1483	0.1560	0.7313	0.6860	0.6376
flchain	Phase 1	15	0.1103	0.2380	0.2180	0.7158	0.6463	0.5568
flchain	Phase 1	30	0.3143	0.5140	0.4930	0.5685	0.3778	0.2480

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Table S4 (continued)

Dataset	Phase	ε	FPR (α=0.01)	FPR (α=0.05)	FPR (α=0.10)	LSR (α=0.01)	LSR (α=0.05)	LSR (α=0.10)
flchain	Phase 1	60	0.5097	0.6123	0.5680	0.2430	0.2070	0.1452
flchain	Phase 1	100	0.5513	0.6190	0.4635	0.1935	0.0948	0.0418
flchain	Phase 1	250	0.3353	0.3594	0.0816	0.0636	0.0013	0.0000
flchain	Phase 1	1000	0.0973	0.2910	0.0000	0.0000	0.0000	0.0044
flchain	Phase 1	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
flchain	Phase 2	0.1	0.0123	0.0513	0.1025	0.9895	0.9535	0.9084
flchain	Phase 2	0.5	0.0080	0.0413	0.0900	0.9905	0.9518	0.9062
flchain	Phase 2	1	0.0107	0.0510	0.0955	0.9913	0.9550	0.9026
flchain	Phase 2	2	0.0097	0.0490	0.1095	0.9895	0.9453	0.9012
flchain	Phase 2	3	0.0093	0.0440	0.0885	0.9873	0.9373	0.8906
flchain	Phase 2	5	0.0103	0.0497	0.0970	0.9700	0.9075	0.8510
flchain	Phase 2	7	0.0100	0.0537	0.1035	0.8933	0.8053	0.7706
flchain	Phase 2	10	0.0160	0.0690	0.1130	0.7548	0.7133	0.7094
flchain	Phase 2	15	0.0360	0.1030	0.1325	0.7385	0.6975	0.6706
flchain	Phase 2	30	0.1827	0.3330	0.2920	0.6783	0.5655	0.4302
flchain	Phase 2	60	0.4463	0.6100	0.6190	0.2970	0.2508	0.1878
flchain	Phase 2	100	0.5327	0.6070	0.4670	0.2345	0.1825	0.1156
flchain	Phase 2	250	0.3690	0.4318	0.2262	0.1124	0.0123	0.0010
flchain	Phase 2	1000	0.1830	0.3097	0.0000	0.0000	0.0000	0.0026
flchain	Phase 2	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
flchain	Phase 3	0.1	0.0097	0.0455	0.0885	0.9903	0.9503	0.9004
flchain	Phase 3	0.5	0.0063	0.0365	0.0795	0.9917	0.9497	0.9036
flchain	Phase 3	1	0.0117	0.0580	0.1110	0.9907	0.9453	0.8886
flchain	Phase 3	2	0.0087	0.0480	0.0970	0.9917	0.9553	0.9076
flchain	Phase 3	3	0.0080	0.0440	0.0810	0.9870	0.9397	0.8932
flchain	Phase 3	5	0.0093	0.0440	0.0980	0.9687	0.8983	0.8586
flchain	Phase 3	7	0.0093	0.0545	0.1060	0.8940	0.7937	0.7940
flchain	Phase 3	10	0.0137	0.0565	0.1135	0.6897	0.6487	0.7204
flchain	Phase 3	15	0.0287	0.0810	0.1410	0.6550	0.6227	0.6818
flchain	Phase 3	30	0.1830	0.2125	0.3190	0.5680	0.4407	0.4272
flchain	Phase 3	60	0.4660	0.4835	0.6880	0.0490	0.0107	0.1938
flchain	Phase 3	100	0.5433	0.4270	0.4840	0.0013	0.0000	0.1834
flchain	Phase 3	250	0.3847	0.2020	0.2845	0.0000	0.0000	0.1340
flchain	Phase 3	1000	0.2587	0.0000	0.0000	0.0000	0.0000	0.0520
flchain	Phase 3	Inf	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

S4. Mechanism Selection and Discrete-Time Diagnostics

Table S5 compares the estimation variance of generalized Randomized Response with optimized local protocols (Section III-B of the main manuscript). Table S6 decomposes the baseline discordance between the Cox model and the discrete-time model for the three coefficients excluded from the Phase 3 denominators, and Table S7 presents the self-referential Phase 3 baseline (Section III-D of the main manuscript).

TABLE S5: Estimation variance of generalized Randomized Response (GRR, direct encoding) against optimized local protocols, for every categorical variable and representative ε (406 rows). The GRR dominance condition $k < 3e^{\varepsilon_j} + 2$ holds in 382 of 406 rows. For the covariates and the event indicator, GRR's variance never exceeds that of unary encoding; the least favorable ratio is 0.747. All 24 unfavorable rows are Phase 3 interval indices, where the disadvantage is bounded by $(K-1)/4$ and reaches at most a factor of 3 (2.97 observed).

Dataset	q	Phase	Mechanism target	Role	k	ε	ε_j	$\mathrm{Var}_{\mathrm{GRR}}/n$	$\mathrm{Var}_{\mathrm{OUE}}/n$	Ratio	$3e^{\varepsilon_j}+2$	GRR optimal	ε^*
lung	7	1	sex	Covariate	2	0.1	0.01429	4900	1.96e+04	0.250	5.043	Yes	—
lung	7	1	sex	Covariate	2	0.5	0.07143	195.9	783.7	0.250	5.222	Yes	—
lung	7	1	sex	Covariate	2	1	0.1429	48.92	195.7	0.250	5.461	Yes	—
lung	7	1	sex	Covariate	2	5	0.7143	1.879	7.515	0.250	8.128	Yes	—
lung	7	1	sex	Covariate	2	10	1.429	0.4145	1.658	0.250	14.52	Yes	—
lung	7	1	sex	Covariate	2	30	4.286	0.01415	0.0566	0.250	220	Yes	—
lung	7	1	sex	Covariate	2	100	14.29	6.249e-07	2.5e-06	0.250	4.801e+06	Yes	—
lung	7	2, 3	sex	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
lung	7	2, 3	sex	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
lung	7	2, 3	sex	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
lung	7	2, 3	sex	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
lung	7	2, 3	sex	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
lung	7	2, 3	sex	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
lung	7	2, 3	sex	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
lung	7	2, 3	delta (event indicator)	Event indicator	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
lung	7	3	k* (interval index)	Interval index	8	0.1	0.01111	5.616e+04	3.24e+04	1.733	5.034	No	6.24
lung	7	3	k* (interval index)	Interval index	8	0.5	0.05556	2162	1296	1.669	5.171	No	6.24
lung	7	3	k* (interval index)	Interval index	8	1	0.1111	515.4	323.7	1.592	5.353	No	6.24
lung	7	3	k* (interval index)	Interval index	8	5	0.5556	14.03	12.63	1.111	7.229	No	6.24
lung	7	3	k* (interval index)	Interval index	8	10	1.111	2.177	2.926	0.744	11.11	Yes	6.24
lung	7	3	k* (interval index)	Interval index	8	30	3.333	0.04657	0.1534	0.304	86.09	Yes	6.24
lung	7	3	k* (interval index)	Interval index	8	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	6.24
pbc	17	1	ascites	Covariate	2	0.1	0.005882	2.89e+04	1.156e+05	0.250	5.018	Yes	—
pbc	17	1	ascites	Covariate	2	0.5	0.02941	1156	4624	0.250	5.09	Yes	—
pbc	17	1	ascites	Covariate	2	1	0.05882	288.9	1156	0.250	5.182	Yes	—
pbc	17	1	ascites	Covariate	2	5	0.2941	11.48	45.91	0.250	6.026	Yes	—
pbc	17	1	ascites	Covariate	2	10	0.5882	2.808	11.23	0.250	7.402	Yes	—
pbc	17	1	ascites	Covariate	2	30	1.765	0.2493	0.9972	0.250	19.52	Yes	—
pbc	17	1	ascites	Covariate	2	100	5.882	0.002804	0.01122	0.250	1078	Yes	—
pbc	17	1	edema	Covariate	3	0.1	0.005882	5.763e+04	1.156e+05	0.499	5.018	Yes	—
pbc	17	1	edema	Covariate	3	0.5	0.02941	2278	4624	0.493	5.09	Yes	—
pbc	17	1	edema	Covariate	3	1	0.05882	561.3	1156	0.486	5.182	Yes	—
pbc	17	1	edema	Covariate	3	5	0.2941	20.03	45.91	0.436	6.026	Yes	—
pbc	17	1	edema	Covariate	3	10	0.5882	4.367	11.23	0.389	7.402	Yes	—

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Table S5 (continued)

Dataset	q	Phase	Mechanism target	Role	k	ε	ε_j	$\mathrm{Var}_{\mathrm{GRR}}/n$	$\mathrm{Var}_{\mathrm{OUE}}/n$	Ratio	$3e^{\varepsilon_j}+2$	GRR optimal	ε^*
pbc	17	1	edema	Covariate	3	30	1.765	0.292	0.9972	0.293	19.52	Yes	—
pbc	17	1	edema	Covariate	3	100	5.882	0.002812	0.01122	0.251	1078	Yes	—
pbc	17	1	hepato	Covariate	2	0.1	0.005882	2.89e+04	1.156e+05	0.250	5.018	Yes	—
pbc	17	1	hepato	Covariate	2	0.5	0.02941	1156	4624	0.250	5.09	Yes	—
pbc	17	1	hepato	Covariate	2	1	0.05882	288.9	1156	0.250	5.182	Yes	—
pbc	17	1	hepato	Covariate	2	5	0.2941	11.48	45.91	0.250	6.026	Yes	—
pbc	17	1	hepato	Covariate	2	10	0.5882	2.808	11.23	0.250	7.402	Yes	—
pbc	17	1	hepato	Covariate	2	30	1.765	0.2493	0.9972	0.250	19.52	Yes	—
pbc	17	1	hepato	Covariate	2	100	5.882	0.002804	0.01122	0.250	1078	Yes	—
pbc	17	1	sex	Covariate	2	0.1	0.005882	2.89e+04	1.156e+05	0.250	5.018	Yes	—
pbc	17	1	sex	Covariate	2	0.5	0.02941	1156	4624	0.250	5.09	Yes	—
pbc	17	1	sex	Covariate	2	1	0.05882	288.9	1156	0.250	5.182	Yes	—
pbc	17	1	sex	Covariate	2	5	0.2941	11.48	45.91	0.250	6.026	Yes	—
pbc	17	1	sex	Covariate	2	10	0.5882	2.808	11.23	0.250	7.402	Yes	—
pbc	17	1	sex	Covariate	2	30	1.765	0.2493	0.9972	0.250	19.52	Yes	—
pbc	17	1	sex	Covariate	2	100	5.882	0.002804	0.01122	0.250	1078	Yes	—
pbc	17	1	spiders	Covariate	2	0.1	0.005882	2.89e+04	1.156e+05	0.250	5.018	Yes	—
pbc	17	1	spiders	Covariate	2	0.5	0.02941	1156	4624	0.250	5.09	Yes	—
pbc	17	1	spiders	Covariate	2	1	0.05882	288.9	1156	0.250	5.182	Yes	—
pbc	17	1	spiders	Covariate	2	5	0.2941	11.48	45.91	0.250	6.026	Yes	—
pbc	17	1	spiders	Covariate	2	10	0.5882	2.808	11.23	0.250	7.402	Yes	—
pbc	17	1	spiders	Covariate	2	30	1.765	0.2493	0.9972	0.250	19.52	Yes	—
pbc	17	1	spiders	Covariate	2	100	5.882	0.002804	0.01122	0.250	1078	Yes	—
pbc	17	1	stage	Covariate	4	0.1	0.005882	8.636e+04	1.156e+05	0.747	5.018	Yes	—
pbc	17	1	stage	Covariate	4	0.5	0.02941	3401	4624	0.736	5.09	Yes	—
pbc	17	1	stage	Covariate	4	1	0.05882	833.7	1156	0.721	5.182	Yes	—
pbc	17	1	stage	Covariate	4	5	0.2941	28.58	45.91	0.623	6.026	Yes	—
pbc	17	1	stage	Covariate	4	10	0.5882	5.927	11.23	0.528	7.402	Yes	—
pbc	17	1	stage	Covariate	4	30	1.765	0.3347	0.9972	0.336	19.52	Yes	—
pbc	17	1	stage	Covariate	4	100	5.882	0.002819	0.01122	0.251	1078	Yes	—
pbc	17	1	trt	Covariate	2	0.1	0.005882	2.89e+04	1.156e+05	0.250	5.018	Yes	—
pbc	17	1	trt	Covariate	2	0.5	0.02941	1156	4624	0.250	5.09	Yes	—
pbc	17	1	trt	Covariate	2	1	0.05882	288.9	1156	0.250	5.182	Yes	—
pbc	17	1	trt	Covariate	2	5	0.2941	11.48	45.91	0.250	6.026	Yes	—
pbc	17	1	trt	Covariate	2	10	0.5882	2.808	11.23	0.250	7.402	Yes	—
pbc	17	1	trt	Covariate	2	30	1.765	0.2493	0.9972	0.250	19.52	Yes	—
pbc	17	1	trt	Covariate	2	100	5.882	0.002804	0.01122	0.250	1078	Yes	—
pbc	17	2, 3	ascites	Covariate	2	0.1	0.005263	3.61e+04	1.444e+05	0.250	5.016	Yes	—
pbc	17	2, 3	ascites	Covariate	2	0.5	0.02632	1444	5776	0.250	5.08	Yes	—
pbc	17	2, 3	ascites	Covariate	2	1	0.05263	360.9	1444	0.250	5.162	Yes	—
pbc	17	2, 3	ascites	Covariate	2	5	0.2632	14.36	57.43	0.250	5.903	Yes	—
pbc	17	2, 3	ascites	Covariate	2	10	0.5263	3.528	14.11	0.250	7.078	Yes	—
pbc	17	2, 3	ascites	Covariate	2	30	1.579	0.3272	1.309	0.250	16.55	Yes	—
pbc	17	2, 3	ascites	Covariate	2	100	5.263	0.005233	0.02093	0.250	581.3	Yes	—
pbc	17	2, 3	edema	Covariate	3	0.1	0.005263	7.201e+04	1.444e+05	0.499	5.016	Yes	—
pbc	17	2, 3	edema	Covariate	3	0.5	0.02632	2850	5776	0.494	5.08	Yes	—
pbc	17	2, 3	edema	Covariate	3	1	0.05263	703.3	1444	0.487	5.162	Yes	—
pbc	17	2, 3	edema	Covariate	3	5	0.2632	25.39	57.43	0.442	5.903	Yes	—
pbc	17	2, 3	edema	Covariate	3	10	0.5263	5.612	14.11	0.398	7.078	Yes	—
pbc	17	2, 3	edema	Covariate	3	30	1.579	0.3947	1.309	0.302	16.55	Yes	—
pbc	17	2, 3	edema	Covariate	3	100	5.263	0.00526	0.02093	0.251	581.3	Yes	—
pbc	17	2, 3	hepato	Covariate	2	0.1	0.005263	3.61e+04	1.444e+05	0.250	5.016	Yes	—
pbc	17	2, 3	hepato	Covariate	2	0.5	0.02632	1444	5776	0.250	5.08	Yes	—
pbc	17	2, 3	hepato	Covariate	2	1	0.05263	360.9	1444	0.250	5.162	Yes	—
pbc	17	2, 3	hepato	Covariate	2	5	0.2632	14.36	57.43	0.250	5.903	Yes	—
pbc	17	2, 3	hepato	Covariate	2	10	0.5263	3.528	14.11	0.250	7.078	Yes	—
pbc	17	2, 3	hepato	Covariate	2	30	1.579	0.3272	1.309	0.250	16.55	Yes	—
pbc	17	2, 3	hepato	Covariate	2	100	5.263	0.005233	0.02093	0.250	581.3	Yes	—
pbc	17	2, 3	sex	Covariate	2	0.1	0.005263	3.61e+04	1.444e+05	0.250	5.016	Yes	—
pbc	17	2, 3	sex	Covariate	2	0.5	0.02632	1444	5776	0.250	5.08	Yes	—
pbc	17	2, 3	sex	Covariate	2	1	0.05263	360.9	1444	0.250	5.162	Yes	—
pbc	17	2, 3	sex	Covariate	2	5	0.2632	14.36	57.43	0.250	5.903	Yes	—
pbc	17	2, 3	sex	Covariate	2	10	0.5263	3.528	14.11	0.250	7.078	Yes	—
pbc	17	2, 3	sex	Covariate	2	30	1.579	0.3272	1.309	0.250	16.55	Yes	—
pbc	17	2, 3	sex	Covariate	2	100	5.263	0.005233	0.02093	0.250	581.3	Yes	—
pbc	17	2, 3	spiders	Covariate	2	0.1	0.005263	3.61e+04	1.444e+05	0.250	5.016	Yes	—
pbc	17	2, 3	spiders	Covariate	2	0.5	0.02632	1444	5776	0.250	5.08	Yes	—
pbc	17	2, 3	spiders	Covariate	2	1	0.05263	360.9	1444	0.250	5.162	Yes	—
pbc	17	2, 3	spiders	Covariate	2	5	0.2632	14.36	57.43	0.250	5.903	Yes	—
pbc	17	2, 3	spiders	Covariate	2	10	0.5263	3.528	14.11	0.250	7.078	Yes	—
pbc	17	2, 3	spiders	Covariate	2	30	1.579	0.3272	1.309	0.250	16.55	Yes	—
pbc	17	2, 3	spiders	Covariate	2	100	5.263	0.005233	0.02093	0.250	581.3	Yes	—
pbc	17	2, 3	stage	Covariate	4	0.1	0.005263	1.079e+05	1.444e+05	0.747	5.016	Yes	—
pbc	17	2, 3	stage	Covariate	4	0.5	0.02632	4257	5776	0.737	5.08	Yes	—
pbc	17	2, 3	stage	Covariate	4	1	0.05263	1046	1444	0.724	5.162	Yes	—
pbc	17	2, 3	stage	Covariate	4	5	0.2632	36.43	57.43	0.634	5.903	Yes	—
pbc	17	2, 3	stage	Covariate	4	10	0.5263	7.696	14.11	0.545	7.078	Yes	—
pbc	17	2, 3	stage	Covariate	4	30	1.579	0.4622	1.309	0.353	16.55	Yes	—
pbc	17	2, 3	stage	Covariate	4	100	5.263	0.005287	0.02093	0.253	581.3	Yes	—
pbc	17	2, 3	trt	Covariate	2	0.1	0.005263	3.61e+04	1.444e+05	0.250	5.016	Yes	—
pbc	17	2, 3	trt	Covariate	2	0.5	0.02632	1444	5776	0.250	5.08	Yes	—
pbc	17	2, 3	trt	Covariate	2	1	0.05263	360.9	1444	0.250	5.162	Yes	—
pbc	17	2, 3	trt	Covariate	2	5	0.2632	14.36	57.43	0.250	5.903	Yes	—
pbc	17	2, 3	trt	Covariate	2	10	0.5263	3.528	14.11	0.250	7.078	Yes	—
pbc	17	2, 3	trt	Covariate	2	30	1.579	0.3272	1.309	0.250	16.55	Yes	—
pbc	17	2, 3	trt	Covariate	2	100	5.263	0.005233	0.02093	0.250	581.3	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	0.1	0.005263	3.61e+04	1.444e+05	0.250	5.016	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	0.5	0.02632	1444	5776	0.250	5.08	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	1	0.05263	360.9	1444	0.250	5.162	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	5	0.2632	14.36	57.43	0.250	5.903	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	10	0.5263	3.528	14.11	0.250	7.078	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	30	1.579	0.3272	1.309	0.250	16.55	Yes	—
pbc	17	2, 3	delta (event indicator)	Event indicator	2	100	5.263	0.005233	0.02093	0.250	581.3	Yes	—
pbc	17	3	k* (interval index)	Interval index	9	0.1	0.005263	2.875e+05	1.444e+05	1.991	5.016	No	16.10
pbc	17	3	k* (interval index)	Interval index	9	0.5	0.02632	1.129e+04	5776	1.955	5.08	No	16.10
pbc	17	3	k* (interval index)	Interval index	9	1	0.05263	2758	1444	1.910	5.162	No	16.10
pbc	17	3	k* (interval index)	Interval index	9	5	0.2632	91.6	57.43	1.595	5.903	No	16.10
pbc	17	3	k* (interval index)	Interval index	9	10	0.5263	18.12	14.11	1.284	7.078	No	16.10
pbc	17	3	k* (interval index)	Interval index	9	30	1.579	0.7995	1.309	0.611	16.55	Yes	16.10
pbc	17	3	k* (interval index)	Interval index	9	100	5.263	0.005423	0.02093	0.259	581.3	Yes	16.10
colon	11	1	adhere	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
colon	11	1	adhere	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
colon	11	1	adhere	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
colon	11	1	adhere	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
colon	11	1	adhere	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
colon	11	1	adhere	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
colon	11	1	adhere	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	differ	Covariate	3	0.1	0.009091	2.409e+04	4.84e+04	0.498	5.027	Yes	—
colon	11	1	differ	Covariate	3	0.5	0.04545	946.3	1936	0.489	5.14	Yes	—
colon	11	1	differ	Covariate	3	1	0.09091	231.3	483.7	0.478	5.286	Yes	—

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Table S5 (continued)

Dataset	q	Phase	Mechanism target	Role	k	ε	ε_j	$\mathrm{Var}_{\mathrm{GRR}}/n$	$\mathrm{Var}_{\mathrm{OUE}}/n$	Ratio	$3e^{\varepsilon_j}+2$	GRR optimal	ε^*
colon	11	1	differ	Covariate	3	5	0.4545	7.777	19.03	0.409	6.726	Yes	—
colon	11	1	differ	Covariate	3	10	0.9091	1.585	4.52	0.351	9.446	Yes	—
colon	11	1	differ	Covariate	3	30	2.727	0.07977	0.2995	0.266	47.87	Yes	—
colon	11	1	differ	Covariate	3	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	extent	Covariate	4	0.1	0.009091	3.608e+04	4.84e+04	0.745	5.027	Yes	—
colon	11	1	extent	Covariate	4	0.5	0.04545	1409	1936	0.728	5.14	Yes	—
colon	11	1	extent	Covariate	4	1	0.09091	341.7	483.7	0.707	5.286	Yes	—
colon	11	1	extent	Covariate	4	5	0.4545	10.8	19.03	0.567	6.726	Yes	—
colon	11	1	extent	Covariate	4	10	0.9091	2.041	4.52	0.451	9.446	Yes	—
colon	11	1	extent	Covariate	4	30	2.727	0.08466	0.2995	0.283	47.87	Yes	—
colon	11	1	extent	Covariate	4	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	node4	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
colon	11	1	node4	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
colon	11	1	node4	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
colon	11	1	node4	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
colon	11	1	node4	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
colon	11	1	node4	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
colon	11	1	node4	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	obstruct	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
colon	11	1	obstruct	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
colon	11	1	obstruct	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
colon	11	1	obstruct	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
colon	11	1	obstruct	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
colon	11	1	obstruct	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
colon	11	1	obstruct	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	perfor	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
colon	11	1	perfor	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
colon	11	1	perfor	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
colon	11	1	perfor	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
colon	11	1	perfor	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
colon	11	1	perfor	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
colon	11	1	perfor	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	rx	Covariate	3	0.1	0.009091	2.409e+04	4.84e+04	0.498	5.027	Yes	—
colon	11	1	rx	Covariate	3	0.5	0.04545	946.3	1936	0.489	5.14	Yes	—
colon	11	1	rx	Covariate	3	1	0.09091	231.3	483.7	0.478	5.286	Yes	—
colon	11	1	rx	Covariate	3	5	0.4545	7.777	19.03	0.409	6.726	Yes	—
colon	11	1	rx	Covariate	3	10	0.9091	1.585	4.52	0.351	9.446	Yes	—
colon	11	1	rx	Covariate	3	30	2.727	0.07977	0.2995	0.266	47.87	Yes	—
colon	11	1	rx	Covariate	3	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	sex	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
colon	11	1	sex	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
colon	11	1	sex	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
colon	11	1	sex	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
colon	11	1	sex	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
colon	11	1	sex	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
colon	11	1	sex	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	1	surg	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
colon	11	1	surg	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
colon	11	1	surg	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
colon	11	1	surg	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
colon	11	1	surg	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
colon	11	1	surg	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
colon	11	1	surg	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
colon	11	2, 3	adhere	Covariate	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—
colon	11	2, 3	adhere	Covariate	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	adhere	Covariate	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	adhere	Covariate	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	adhere	Covariate	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	adhere	Covariate	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	adhere	Covariate	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	2, 3	differ	Covariate	3	0.1	0.007692	3.367e+04	6.76e+04	0.498	5.023	Yes	—
colon	11	2, 3	differ	Covariate	3	0.5	0.03846	1326	2704	0.491	5.118	Yes	—
colon	11	2, 3	differ	Covariate	3	1	0.07692	325.3	675.7	0.481	5.24	Yes	—
colon	11	2, 3	differ	Covariate	3	5	0.3846	11.22	26.71	0.420	6.407	Yes	—
colon	11	2, 3	differ	Covariate	3	10	0.7692	2.355	6.436	0.366	8.474	Yes	—
colon	11	2, 3	differ	Covariate	3	30	2.308	0.1349	0.4908	0.275	32.15	Yes	—
colon	11	2, 3	differ	Covariate	3	100	7.692	0.0004569	0.001827	0.250	6576	Yes	—
colon	11	2, 3	extent	Covariate	4	0.1	0.007692	5.044e+04	6.76e+04	0.746	5.023	Yes	—
colon	11	2, 3	extent	Covariate	4	0.5	0.03846	1977	2704	0.731	5.118	Yes	—
colon	11	2, 3	extent	Covariate	4	1	0.07692	481.7	675.7	0.713	5.24	Yes	—
colon	11	2, 3	extent	Covariate	4	5	0.3846	15.77	26.71	0.590	6.407	Yes	—
colon	11	2, 3	extent	Covariate	4	10	0.7692	3.1	6.436	0.482	8.474	Yes	—
colon	11	2, 3	extent	Covariate	4	30	2.308	0.1471	0.4908	0.300	32.15	Yes	—
colon	11	2, 3	extent	Covariate	4	100	7.692	0.0004572	0.001827	0.250	6576	Yes	—
colon	11	2, 3	node4	Covariate	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—
colon	11	2, 3	node4	Covariate	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	node4	Covariate	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	node4	Covariate	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	node4	Covariate	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	node4	Covariate	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	node4	Covariate	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	2, 3	obstruct	Covariate	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—
colon	11	2, 3	obstruct	Covariate	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	obstruct	Covariate	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	obstruct	Covariate	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	obstruct	Covariate	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	obstruct	Covariate	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	obstruct	Covariate	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	2, 3	perfor	Covariate	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—
colon	11	2, 3	perfor	Covariate	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	perfor	Covariate	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	perfor	Covariate	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	perfor	Covariate	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	perfor	Covariate	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	perfor	Covariate	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	2, 3	rx	Covariate	3	0.1	0.007692	3.367e+04	6.76e+04	0.498	5.023	Yes	—
colon	11	2, 3	rx	Covariate	3	0.5	0.03846	1326	2704	0.491	5.118	Yes	—
colon	11	2, 3	rx	Covariate	3	1	0.07692	325.3	675.7	0.481	5.24	Yes	—
colon	11	2, 3	rx	Covariate	3	5	0.3846	11.22	26.71	0.420	6.407	Yes	—
colon	11	2, 3	rx	Covariate	3	10	0.7692	2.355	6.436	0.366	8.474	Yes	—
colon	11	2, 3	rx	Covariate	3	30	2.308	0.1349	0.4908	0.275	32.15	Yes	—
colon	11	2, 3	rx	Covariate	3	100	7.692	0.0004569	0.001827	0.250	6576	Yes	—
colon	11	2, 3	sex	Covariate	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—
colon	11	2, 3	sex	Covariate	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	sex	Covariate	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	sex	Covariate	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	sex	Covariate	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	sex	Covariate	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	sex	Covariate	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	2, 3	surg	Covariate	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—

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Table S5 (continued)

Dataset	q	Phase	Mechanism target	Role	k	ε	ε_j	$\mathrm{Var}_{\mathrm{GRR}}/n$	$\mathrm{Var}_{\mathrm{OUE}}/n$	Ratio	$3e^{\varepsilon_j}+2$	GRR optimal	ε^*
colon	11	2, 3	surg	Covariate	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	surg	Covariate	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	surg	Covariate	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	surg	Covariate	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	surg	Covariate	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	surg	Covariate	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	0.1	0.007692	1.69e+04	6.76e+04	0.250	5.023	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	0.5	0.03846	675.9	2704	0.250	5.118	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	1	0.07692	168.9	675.7	0.250	5.24	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	5	0.3846	6.677	26.71	0.250	6.407	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	10	0.7692	1.609	6.436	0.250	8.474	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	30	2.308	0.1227	0.4908	0.250	32.15	Yes	—
colon	11	2, 3	delta (event indicator)	Event indicator	2	100	7.692	0.0004567	0.001827	0.250	6576	Yes	—
colon	11	3	k* (interval index)	Interval index	10	0.1	0.007692	1.511e+05	6.76e+04	2.235	5.023	No	12.75
colon	11	3	k* (interval index)	Interval index	10	0.5	0.03846	5879	2704	2.175	5.118	No	12.75
colon	11	3	k* (interval index)	Interval index	10	1	0.07692	1420	675.7	2.102	5.24	No	12.75
colon	11	3	k* (interval index)	Interval index	10	5	0.3846	43.04	26.71	1.611	6.407	No	12.75
colon	11	3	k* (interval index)	Interval index	10	10	0.7692	7.574	6.436	1.177	8.474	No	12.75
colon	11	3	k* (interval index)	Interval index	10	30	2.308	0.2203	0.4908	0.449	32.15	Yes	12.75
colon	11	3	k* (interval index)	Interval index	10	100	7.692	0.0004584	0.001827	0.251	6576	Yes	12.75
rotterdam	9	1	chemo	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
rotterdam	9	1	chemo	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
rotterdam	9	1	chemo	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
rotterdam	9	1	chemo	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
rotterdam	9	1	chemo	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
rotterdam	9	1	chemo	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
rotterdam	9	1	chemo	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
rotterdam	9	1	grade	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
rotterdam	9	1	grade	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
rotterdam	9	1	grade	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
rotterdam	9	1	grade	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
rotterdam	9	1	grade	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
rotterdam	9	1	grade	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
rotterdam	9	1	grade	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
rotterdam	9	1	hormon	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
rotterdam	9	1	hormon	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
rotterdam	9	1	hormon	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
rotterdam	9	1	hormon	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
rotterdam	9	1	hormon	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
rotterdam	9	1	hormon	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
rotterdam	9	1	hormon	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
rotterdam	9	1	meno	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
rotterdam	9	1	meno	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
rotterdam	9	1	meno	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
rotterdam	9	1	meno	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
rotterdam	9	1	meno	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
rotterdam	9	1	meno	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
rotterdam	9	1	meno	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
rotterdam	9	1	size	Covariate	3	0.1	0.01111	1.611e+04	3.24e+04	0.497	5.034	Yes	—
rotterdam	9	1	size	Covariate	3	0.5	0.05556	630.3	1296	0.486	5.171	Yes	—
rotterdam	9	1	size	Covariate	3	1	0.1111	153.3	323.7	0.474	5.353	Yes	—
rotterdam	9	1	size	Covariate	3	5	0.5556	4.97	12.63	0.393	7.229	Yes	—
rotterdam	9	1	size	Covariate	3	10	1.111	0.9724	2.926	0.332	11.11	Yes	—
rotterdam	9	1	size	Covariate	3	30	3.333	0.03973	0.1534	0.259	86.09	Yes	—
rotterdam	9	1	size	Covariate	3	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
rotterdam	9	2, 3	chemo	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
rotterdam	9	2, 3	grade	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
rotterdam	9	2, 3	hormon	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
rotterdam	9	2, 3	meno	Covariate	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
rotterdam	9	2, 3	size	Covariate	3	0.1	0.009091	2.409e+04	4.84e+04	0.498	5.027	Yes	—
rotterdam	9	2, 3	size	Covariate	3	0.5	0.04545	946.3	1936	0.489	5.14	Yes	—
rotterdam	9	2, 3	size	Covariate	3	1	0.09091	231.3	483.7	0.478	5.286	Yes	—
rotterdam	9	2, 3	size	Covariate	3	5	0.4545	7.777	19.03	0.409	6.726	Yes	—
rotterdam	9	2, 3	size	Covariate	3	10	0.9091	1.585	4.52	0.351	9.446	Yes	—
rotterdam	9	2, 3	size	Covariate	3	30	2.727	0.07977	0.2995	0.266	47.87	Yes	—
rotterdam	9	2, 3	size	Covariate	3	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	0.1	0.009091	1.21e+04	4.84e+04	0.250	5.027	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	0.5	0.04545	483.9	1936	0.250	5.14	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	1	0.09091	120.9	483.7	0.250	5.286	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	5	0.4545	4.758	19.03	0.250	6.726	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	10	0.9091	1.13	4.52	0.250	9.446	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	30	2.727	0.07487	0.2995	0.250	47.87	Yes	—
rotterdam	9	2, 3	delta (event indicator)	Event indicator	2	100	9.091	0.0001127	0.0004508	0.250	2.662e+04	Yes	—
rotterdam	9	3	k* (interval index)	Interval index	12	0.1	0.009091	1.32e+05	4.84e+04	2.727	5.027	No	13.24
rotterdam	9	3	k* (interval index)	Interval index	12	0.5	0.04545	5108	1936	2.639	5.14	No	13.24
rotterdam	9	3	k* (interval index)	Interval index	12	1	0.09091	1225	483.7	2.533	5.286	No	13.24
rotterdam	9	3	k* (interval index)	Interval index	12	5	0.4545	34.96	19.03	1.837	6.726	No	13.24
rotterdam	9	3	k* (interval index)	Interval index	12	10	0.9091	5.683	4.52	1.257	9.446	No	13.24
rotterdam	9	3	k* (interval index)	Interval index	12	30	2.727	0.1238	0.2995	0.413	47.87	Yes	13.24
rotterdam	9	3	k* (interval index)	Interval index	12	100	9.091	0.0001128	0.0004508	0.250	2.662e+04	Yes	13.24
flchain	7	1	mgus	Covariate	2	0.1	0.01429	4900	1.96e+04	0.250	5.043	Yes	—
flchain	7	1	mgus	Covariate	2	0.5	0.07143	195.9	783.7	0.250	5.222	Yes	—
flchain	7	1	mgus	Covariate	2	1	0.1429	48.92	195.7	0.250	5.461	Yes	—
flchain	7	1	mgus	Covariate	2	5	0.7143	1.879	7.515	0.250	8.128	Yes	—
flchain	7	1	mgus	Covariate	2	10	1.429	0.4145	1.658	0.250	14.52	Yes	—
flchain	7	1	mgus	Covariate	2	30	4.286	0.01415	0.0566	0.250	220	Yes	—

continued on next page

Table S5 (continued)

Dataset	q	Phase	Mechanism target	Role	k	ε	ε_j	$\mathrm{Var}_{\mathrm{GRR}}/n$	$\mathrm{Var}_{\mathrm{OUE}}/n$	Ratio	$3e^{\varepsilon_j}+2$	GRR optimal	ε^*
flchain	7	1	mgus	Covariate	2	100	14.29	6.249e-07	2.5e-06	0.250	4.801e+06	Yes	—
flchain	7	1	sex	Covariate	2	0.1	0.01429	4900	1.96e+04	0.250	5.043	Yes	—
flchain	7	1	sex	Covariate	2	0.5	0.07143	195.9	783.7	0.250	5.222	Yes	—
flchain	7	1	sex	Covariate	2	1	0.1429	48.92	195.7	0.250	5.461	Yes	—
flchain	7	1	sex	Covariate	2	5	0.7143	1.879	7.515	0.250	8.128	Yes	—
flchain	7	1	sex	Covariate	2	10	1.429	0.4145	1.658	0.250	14.52	Yes	—
flchain	7	1	sex	Covariate	2	30	4.286	0.01415	0.0566	0.250	220	Yes	—
flchain	7	1	sex	Covariate	2	100	14.29	6.249e-07	2.5e-06	0.250	4.801e+06	Yes	—
flchain	7	2, 3	mgus	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
flchain	7	2, 3	mgus	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
flchain	7	2, 3	mgus	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
flchain	7	2, 3	mgus	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
flchain	7	2, 3	mgus	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
flchain	7	2, 3	mgus	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
flchain	7	2, 3	mgus	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
flchain	7	2, 3	sex	Covariate	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
flchain	7	2, 3	sex	Covariate	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
flchain	7	2, 3	sex	Covariate	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
flchain	7	2, 3	sex	Covariate	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
flchain	7	2, 3	sex	Covariate	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
flchain	7	2, 3	sex	Covariate	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
flchain	7	2, 3	sex	Covariate	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	0.1	0.01111	8100	3.24e+04	0.250	5.034	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	0.5	0.05556	323.9	1296	0.250	5.171	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	1	0.1111	80.92	323.7	0.250	5.353	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	5	0.5556	3.158	12.63	0.250	7.229	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	10	1.111	0.7316	2.926	0.250	11.11	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	30	3.333	0.03836	0.1534	0.250	86.09	Yes	—
flchain	7	2, 3	delta (event indicator)	Event indicator	2	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	—
flchain	7	3	k* (interval index)	Interval index	13	0.1	0.01111	9.621e+04	3.24e+04	2.970	5.034	No	11.69
flchain	7	3	k* (interval index)	Interval index	13	0.5	0.05556	3694	1296	2.851	5.171	No	11.69
flchain	7	3	k* (interval index)	Interval index	13	1	0.1111	877.4	323.7	2.711	5.353	No	11.69
flchain	7	3	k* (interval index)	Interval index	13	5	0.5556	23.09	12.63	1.828	7.229	No	11.69
flchain	7	3	k* (interval index)	Interval index	13	10	1.111	3.381	2.926	1.155	11.11	No	11.69
flchain	7	3	k* (interval index)	Interval index	13	30	3.333	0.05342	0.1534	0.348	86.09	Yes	11.69
flchain	7	3	k* (interval index)	Interval index	13	100	11.11	1.495e-05	5.978e-05	0.250	2.007e+05	Yes	11.69

TABLE S6

BASELINE DISCORDANCE BETWEEN THE COX MODEL AND THE DISCRETE-TIME GLM, FOR THE THREE COEFFICIENTS CLASSIFIED DIFFERENTLY AT $\varepsilon = \infty$. THE CAUSES DIFFER: FOR FLCHAIN `SAMPLE.YR` THE DOMINANT CONTRIBUTION IS TIME DISCRETIZATION ($+72.5\%$, AGAINST $+0.5\%$ FROM THE LINK FUNCTION); FOR LUNG `PH.KARNO` THE LINK DOMINATES ($+28.9\%$); FOR FLCHAIN `KAPPA` THE LINK CONTRIBUTES 21.7%, BUT THE CLASSIFICATION CHANGES ONLY BECAUSE THE COX p-VALUE SITS AT 1.06α. PERCENTAGES ARE THE RELATIVE CHANGE IN $|z|$ ATTRIBUTABLE TO THE STEP NAMED (TIME DISCRETIZATION: EXACT-TIME AGAINST GROUPED-TIME COX; LINK: CLOGLOG AGAINST LOGIT), STATED AS THE LARGER $|z|$ OVER THE SMALLER MINUS ONE AND MARKED $+$ WHERE THE STEP LOWERS $|z|$; FOR `KAPPA` THE LINK STEP RAISES IT. THE STEPS COMPOSE MULTIPLICATIVELY (ADDITIVELY IN $\log|z|$); THEREFORE, THE FOUR PERCENTAGES OF A COEFFICIENT DO NOT SUM TO ITS TOTAL.

Dataset	Variable	Model	K	Beta	SE	z	p	Sig
flchain	kappa	D. Discrete-time, logit, spline baseline (Phase 3)	13	0.0773	0.0365	2.116	0.0343	Yes
flchain	kappa	C. Discrete-time, logit, saturated baseline	13	0.0773	0.0365	2.118	0.0342	Yes
flchain	kappa	B. Discrete-time, cloglog, saturated baseline	13	0.0569	0.0327	1.740	0.0818	No
flchain	kappa	A'. Cox, times grouped into K intervals	13	0.0584	0.0314	1.862	0.0627	No
flchain	kappa	A. Cox, exact times	–	0.0605	0.0313	1.933	0.0532	No
flchain	kappa	D at K = 11	11	0.0820	0.0374	2.193	0.0283	Yes
flchain	kappa	D at K = 15	15	0.0730	0.0360	2.029	0.0424	Yes
flchain	kappa	D at K = 100 (largest swept)	100	0.0625	0.0321	1.949	0.0513	No
lung	ph.karno	D. Discrete-time, logit, spline baseline (Phase 3)	8	0.0201	0.0139	1.446	0.1480	No
lung	ph.karno	C. Discrete-time, logit, saturated baseline	8	0.0206	0.0140	1.472	0.1410	No
lung	ph.karno	B. Discrete-time, cloglog, saturated baseline	8	0.0223	0.0118	1.898	0.0577	No
lung	ph.karno	A'. Cox, times grouped into K intervals	8	0.0211	0.0112	1.888	0.0590	No
lung	ph.karno	A. Cox, exact times	–	0.0225	0.0112	1.998	0.0457	Yes
lung	ph.karno	D at K = 6	6	0.0221	0.0146	1.514	0.1300	No
lung	ph.karno	D at K = 10	10	0.0196	0.0134	1.460	0.1443	No
lung	ph.karno	D at K = 99 (largest swept)	99	0.0220	0.0114	1.929	0.0538	No
flchain	sample.yr	D. Discrete-time, logit, spline baseline (Phase 3)	13	0.0360	0.0199	1.810	0.0703	No
flchain	sample.yr	C. Discrete-time, logit, saturated baseline	13	0.0363	0.0199	1.823	0.0683	No
flchain	sample.yr	B. Discrete-time, cloglog, saturated baseline	13	0.0352	0.0192	1.832	0.0670	No
flchain	sample.yr	A'. Cox, times grouped into K intervals	13	0.0351	0.0192	1.824	0.0681	No
flchain	sample.yr	A. Cox, exact times	–	0.0604	0.0192	3.146	0.0017	Yes
flchain	sample.yr	D at K = 11	11	0.0307	0.0200	1.539	0.1238	No
flchain	sample.yr	D at K = 15	15	0.0380	0.0198	1.917	0.0552	No
flchain	sample.yr	D at K = 100 (largest swept)	100	0.0541	0.0193	2.800	0.0051	Yes

TABLE S7

THE SELF-REFERENTIAL PHASE 3 BASELINE. "SUBMITTED": THE DEFINITION USED IN THE MAIN MANUSCRIPT (COX-REFERENCED, THREE COEFFICIENTS EXCLUDED); "NO_EXCLUSION": THE SAME DEFINITION WITHOUT THE EXCLUSION; "SELF_REFERENTIAL": LSR AND FPR DEFINED AGAINST PHASE 3'S OWN $\varepsilon = \infty$ FIT, USING ALL 53 COEFFICIENTS. THE SELF-REFERENTIAL DEFINITION REPRODUCES THE THRESHOLDS OF THE MAIN MANUSCRIPT IN 29 OF 30 CASES (22 OF 30 IF THE EXCLUSION IS SIMPLY DROPPED); THE ONE EXCEPTION IS THE RIGHT-TAIL FPR THRESHOLD OF LUNG UNDER TRAIN/TEST (FROM 0.1 TO BEYOND THE GRID). A VALUE OF 0.1, THE SMALLEST GRID POINT, MEANS THAT THE CRITERION IS MET AT EVERY BUDGET (SHOWN AS — IN TABLE V OF THE MAIN MANUSCRIPT).

arm	dataset	threshold	submitted	no_exclusion	self_referential	same_without_exclusion	same_self_referential
in-sample	lung	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	60	60	60	TRUE	TRUE
in-sample	lung	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	250	>1000	250	FALSE	TRUE
in-sample	lung	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	0.1	0.1	0.1	TRUE	TRUE
in-sample	pbc	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	100	100	100	TRUE	TRUE
in-sample	pbc	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	1000	1000	1000	TRUE	TRUE
in-sample	pbc	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	250	250	250	TRUE	TRUE
in-sample	colon	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	30	30	30	TRUE	TRUE
in-sample	colon	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	1000	1000	1000	TRUE	TRUE
in-sample	colon	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	250	250	250	TRUE	TRUE
in-sample	rotterdam	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	30	30	30	TRUE	TRUE
in-sample	rotterdam	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	250	250	250	TRUE	TRUE
in-sample	rotterdam	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	60	60	60	TRUE	TRUE
in-sample	flchain	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	30	60	30	FALSE	TRUE
in-sample	flchain	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	60	>1000	60	FALSE	TRUE
in-sample	flchain	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	1000	>1000	1000	FALSE	TRUE
train/test	lung	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	60	250	60	FALSE	TRUE
train/test	lung	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	>1000	>1000	>1000	TRUE	TRUE
train/test	lung	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	0.1	0.1	>1000	TRUE	FALSE
train/test	pbc	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	250	250	250	TRUE	TRUE
train/test	pbc	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	>1000	>1000	>1000	TRUE	TRUE
train/test	pbc	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	1000	1000	1000	TRUE	TRUE
train/test	colon	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	60	60	60	TRUE	TRUE
train/test	colon	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	>1000	>1000	>1000	TRUE	TRUE
train/test	colon	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	1000	1000	1000	TRUE	TRUE
train/test	rotterdam	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	30	30	30	TRUE	TRUE
train/test	rotterdam	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	1000	1000	1000	TRUE	TRUE
train/test	rotterdam	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	>1000	>1000	>1000	TRUE	TRUE
train/test	flchain	$\varepsilon^*_{\mathrm{LSR}\leq 0.50}$	30	60	30	FALSE	TRUE
train/test	flchain	$\varepsilon^*_{\mathrm{LSR}\leq 0.10}$	60	>1000	60	FALSE	TRUE
train/test	flchain	$\varepsilon^*_{\mathrm{FPR}\leq 0.10}$	1000	>1000	1000	FALSE	TRUE

S5. CLIPPING BOUNDS AND THE FORMAL COMPARISON ARM

Table S8 compares the three clipping-bound strategies of Section III-I of the main manuscript at representative budgets.

TABLE S8

THE THREE CLIPPING-BOUND STRATEGIES AT REPRESENTATIVE BUDGETS: MEAN LSR, MEAN IN-SAMPLE C-INDEX, IN-SAMPLE ΔC (AGAINST THE ARM'S OWN $\varepsilon = \infty$; THE ΔC OF TABLE V OF THE MAIN MANUSCRIPT IS INSTEAD COMPUTED ON THE TEST C-INDEX), AND MEAN TEST C-INDEX FOR DATA-DRIVEN (DD), FIXED, AND WIDENED (WID.) BOUNDS. ONLY FIXED MAKES THE COVARIATE PERTURBATION OF PHASE 1 FORMALLY ε-LOCAL-DP. AT THE UNTABULATED $\varepsilon = 250$ THE FIXED-BOUND MEAN LSR OF ROTTERDAM (0.073) LIES BELOW THE DATA-DRIVEN VALUE (0.101), THE ONLY BUDGET AT WHICH IT DOES SO BY MORE THAN MONTE-CARLO ERROR (AT $\varepsilon = 0.5$ THE TWO DIFFER BY 0.003); THE THRESHOLDS OF ALL THREE STRATEGIES UNDER THE FOUR CRITERIA OF TABLE V OF THE MAIN MANUSCRIPT ARE TABLE S10.

		Mean LSR			In-sample C-index			ΔC (in-sample)			Test C-index		
Dataset	ε	dd	fixed	wid.	dd	fixed	wid.	dd	fixed	wid.	dd	fixed	wid.
lung	1	0.9363	0.9450	0.9377	0.5683	0.5677	0.5681	0.0829	0.0835	0.0831	0.5122	0.5087	0.5140
lung	5	0.8957	0.9090	0.9080	0.5759	0.5740	0.5731	0.0753	0.0772	0.0780	0.5581	0.5434	0.5445
lung	10	0.8077	0.8453	0.8340	0.5927	0.5853	0.5846	0.0585	0.0659	0.0665	0.5944	0.5780	0.5756
lung	30	0.5323	0.5597	0.5960	0.6322	0.6197	0.6156	0.0190	0.0314	0.0356	0.6177	0.6153	0.6118
lung	100	0.3430	0.4337	0.4810	0.6471	0.6447	0.6414	0.0040	0.0064	0.0098	0.6178	0.6214	0.6200
lung	1000	0.1243	0.1540	0.1847	0.6511	0.6512	0.6508	0.0000	-0.0000	0.0004	0.6141	0.6144	0.6140
lung	∞	0.0000	0.0000	0.0000	0.6512	0.6512	0.6512	0.0000	0.0000	0.0000	0.6140	0.6140	0.6140
pbc	1	0.9381	0.9436	0.9371	0.6047	0.6036	0.6047	0.2522	0.2533	0.2522	0.5318	0.5352	0.5274
pbc	5	0.9228	0.9339	0.9310	0.6135	0.6111	0.6105	0.2435	0.2459	0.2464	0.6424	0.6255	0.6328
pbc	10	0.8831	0.9107	0.9042	0.6344	0.6286	0.6297	0.2226	0.2283	0.2272	0.7306	0.7094	0.7118
pbc	30	0.6385	0.7383	0.7310	0.7319	0.7201	0.7220	0.1250	0.1368	0.1350	0.7946	0.7727	0.7767
pbc	100	0.3616	0.5760	0.5385	0.8219	0.8021	0.8035	0.0351	0.0548	0.0534	0.8236	0.7967	0.7997
pbc	1000	0.0134	0.3126	0.0853	0.8556	0.8450	0.8525	0.0014	0.0119	0.0044	0.8310	0.8290	0.8318
pbc	∞	0.0000	0.0000	0.0000	0.8569	0.8569	0.8569	0.0000	0.0000	0.0000	0.8309	0.8309	0.8309
colon	1	0.9446	0.9482	0.9384	0.5448	0.5449	0.5451	0.1262	0.1262	0.1260	0.5185	0.5178	0.5173
colon	5	0.8494	0.8550	0.8532	0.5534	0.5533	0.5533	0.1177	0.1177	0.1178	0.5792	0.5802	0.5795
colon	10	0.6856	0.6901	0.6862	0.5722	0.5720	0.5724	0.0989	0.0990	0.0987	0.6188	0.6192	0.6187
colon	30	0.3054	0.3272	0.3306	0.6393	0.6393	0.6393	0.0318	0.0317	0.0318	0.6519	0.6512	0.6522
colon	100	0.1548	0.1820	0.1778	0.6689	0.6685	0.6687	0.0022	0.0026	0.0024	0.6577	0.6569	0.6570
colon	1000	0.0024	0.0620	0.0438	0.6710	0.6704	0.6706	0.0000	0.0007	0.0005	0.6588	0.6586	0.6587
colon	∞	0.0000	0.0000	0.0000	0.6711	0.6711	0.6711	0.0000	0.0000	0.0000	0.6588	0.6588	0.6588
rotterdam	1	0.9345	0.9463	0.9430	0.5234	0.5230	0.5229	0.1556	0.1560	0.1561	0.5329	0.5307	0.5317
rotterdam	5	0.6710	0.6875	0.6778	0.5360	0.5352	0.5357	0.1430	0.1438	0.1433	0.6057	0.5967	0.5985
rotterdam	10	0.3953	0.4670	0.4535	0.5595	0.5574	0.5578	0.1194	0.1215	0.1211	0.6250	0.6130	0.6153
rotterdam	30	0.3017	0.4092	0.3620	0.6197	0.6140	0.6147	0.0593	0.0649	0.0643	0.6418	0.6265	0.6300
rotterdam	100	0.1813	0.2188	0.2670	0.6530	0.6348	0.6368	0.0260	0.0441	0.0422	0.6653	0.6443	0.6470
rotterdam	1000	0.0305	0.0488	0.0715	0.6778	0.6746	0.6762	0.0012	0.0044	0.0028	0.6748	0.6740	0.6745
rotterdam	∞	0.0000	0.0000	0.0000	0.6790	0.6792	0.6790	0.0000	-0.0002	0.0000	0.6749	0.6749	0.6749
flchain	1	0.8985	0.9113	0.9305	0.5186	0.5179	0.5174	0.2716	0.2724	0.2729	0.6276	0.6091	0.5584
flchain	5	0.7075	0.7117	0.7485	0.5430	0.5371	0.5260	0.2473	0.2531	0.2643	0.7603	0.7446	0.6905
flchain	10	0.6860	0.6912	0.7037	0.5753	0.5649	0.5420	0.2149	0.2253	0.2482	0.7769	0.7700	0.7393
flchain	30	0.3777	0.5250	0.6595	0.6700	0.6532	0.6031	0.1203	0.1371	0.1872	0.7831	0.7800	0.7779
flchain	100	0.0948	0.0995	0.2408	0.7632	0.7533	0.7176	0.0271	0.0369	0.0727	0.7884	0.7855	0.7840
flchain	1000	0.0000	0.0000	0.0000	0.7898	0.7895	0.7885	0.0004	0.0008	0.0018	0.7905	0.7905	0.7904
flchain	∞	0.0000	0.0000	0.0000	0.7903	0.7903	0.7903	0.0000	0.0000	0.0000	0.7905	0.7905	0.7905

Table S9 presents the fixed bound adopted for each of the 51 covariates, with its tier and the external source value from which it was derived. The applied clause, the unit conversion, and the full source citation are recorded in `tabS9_fixed_bounds.csv` in this repository, and the clipping rate in `docs/fixed_bounds_clipping_check.csv`; they are omitted from the table below only because they do not fit the page. The protocol text itself, with its six tiers and their clauses, is `docs/fixed_bounds_source_protocol.md` in the same repository. The rejected candidate values, with the reason each was rejected, are in the same location and support the disclosures of Section S1 of this document.

TABLE S9: Fixed clipping bounds for all 51 covariates, with tier, adopted domain, and the source value and access date from which the bound was derived.

Dataset	Variable	Tier	Adopted domain	Unit	Source value	Unit	Accessed
lung	sex	0	{1, 2}	categorical	1, 2	categorical	2026-08-25
pbc	ascites	0	{0, 1}	binary	0, 1	binary	2026-08-25
pbc	edema	0	{0, 0.5, 1}	categorical	0, 0.5, 1	categorical	2026-08-25
pbc	hepato	0	{0, 1}	binary	0, 1	binary	2026-08-25
pbc	sex	0	{f, m}	categorical	f, m	categorical	2026-08-25
pbc	spiders	0	{0, 1}	binary	0, 1	binary	2026-08-25
pbc	stage	0	{1, 2, 3, 4}	categorical	1, 2, 3, 4	categorical	2026-08-25
pbc	trt	0	{1, 2}	categorical	1, 2	categorical	2026-08-25
colon	adhere	0	{0, 1}	binary	0, 1	binary	2026-08-25
colon	differ	0	{1, 2, 3}	categorical	1, 2, 3	categorical	2026-08-25
colon	extent	0	{1, 2, 3, 4}	categorical	1, 2, 3, 4	categorical	2026-08-25
colon	node4	0	{0, 1}	binary	0, 1	binary	2026-08-25
colon	obstruct	0	{0, 1}	binary	0, 1	binary	2026-08-25
colon	perfor	0	{0, 1}	binary	0, 1	binary	2026-08-25
colon	rx	0	{Obs, Lev, Lev+5FU}	categorical	Obs, Lev, Lev+5FU	categorical	2026-08-25
colon	sex	0	{0, 1}	binary	0, 1	binary	2026-08-25
colon	surg	0	{0, 1}	binary	0, 1	binary	2026-08-25

Table S9, continued

Dataset	Variable	Tier	Adopted domain	Unit	Source value	Unit	Accessed
rotterdam	chemo	0	{0, 1}	binary	0, 1	binary	2026-08-25
rotterdam	grade	0	{2, 3}	categorical	2, 3	categorical	2026-08-25
rotterdam	hormon	0	{0, 1}	binary	0, 1	binary	2026-08-25
rotterdam	meno	0	{0, 1}	binary	0, 1	binary	2026-08-25
rotterdam	size	0	{≤20, 20-50, >50}	categorical	≤20, 20-50, >50	categorical	2026-08-25
flchain	mgus	0	{0, 1}	binary	0, 1	binary	2026-08-25
flchain	sex	0	{F, M}	categorical	F, M	categorical	2026-08-25
lung	pat.karno	1	[0, 100]	score	0-100	score	2026-08-15
lung	ph.ecog	1	[0, 5]	score	0-5	score	2026-08-15
lung	ph.karno	1	[0, 100]	score	0-100	score	2026-08-15
pbc	albumin	2	[0, 8]	g/dL	8	g/dL	2026-08-23
pbc	alk.phos	2	[0, 20000]	U/L	14589	IU/L	2026-08-23
pbc	ast	2	[0, 10000]	U/L	10000	IU/L	2026-08-23
pbc	bili	2	[0, 50]	mg/dL	41	mg/dL	2026-08-23
pbc	chol	2	[0, 2000]	mg/dL	1730	mg/dL	2026-08-23
pbc	copper	2	[0, 4000]	ug/24h	3080	ug/24h	2026-08-30
pbc	platelet	2	[0, 6000]	x10^9/L	6	x10^6/mm^3	2026-08-23
pbc	protime	2	[0, 200]	seconds	>180	seconds	2026-08-23
pbc	trig	2	[0, 2000]	mg/dL	2000	mg/dL	2026-08-23
flchain	creatinine	2	[0, 80]	mg/dL	73.8	mg/dL	2026-08-23
flchain	kappa	2	[0, 30]	mg/dL	281.8	mg/L	2026-08-23
flchain	lambda	2	[0, 40]	mg/dL	312.6	mg/L	2026-08-23
colon	nodes	3	[0, 90]	count	90	count	2026-08-15
rotterdam	nodes	3	[0, 90]	count	90	count	2026-08-15
lung	age	4	[18, 110]	years	18-110	years	2026-08-23
pbc	age	4	[18, 110]	years	18-110	years	2026-08-23
colon	age	4	[18, 110]	years	18-110	years	2026-08-23
rotterdam	age	4	[18, 110]	years	18-110	years	2026-08-23
flchain	age	4	[50, 110]	years	50	years	2026-08-23
lung	meal.cal	5	[0, 4000]	kcal/day	3720	kcal/day	2026-08-15
lung	wt.loss	5	[-50, 100]	lb/6 months	32.9 (loss) / 11.35 (gain)	kg	2026-08-15
rotterdam	er	5	[0, 2000]	fmol/mg	1000-1102	fmol/mg	2026-08-15
rotterdam	pgr	5	[0, 2000]	fmol/mg	1000-1102	fmol/mg	2026-08-15
flchain	sample.yr	5	[1995, 2003]	calendar year	1995-03-13 to 2003-11-21	date	2026-08-23

Table S10 gives the reference thresholds under each strategy, and Fig. S6 the formal comparison arm of Section IV-I of the main manuscript.

TABLE S10

REFERENCE ε VALUES OF TABLE V OF THE MAIN MANUSCRIPT UNDER ALL THREE CLIPPING-BOUND STRATEGIES (PHASE 1, FULL 15-POINT GRID): DATA-DRIVEN (DD), FIXED, AND WIDENED (WID.). CRITERIA AS IN SECTION III-G OF THE MAIN MANUSCRIPT, ΔC AGAINST THE ARM'S OWN $\varepsilon = \infty$. THE DD COLUMNS REPRODUCE THE PHASE 1 ROWS OF TABLE V. AT $\Delta C \leq 0.05$, THE FIXED-BOUND VALUE FOR PBC IS 60, WHEREAS THE TEST C-INDEX COLUMNS OF TABLE S8, WHICH TABULATES REPRESENTATIVE BUDGETS ONLY, WOULD SUGGEST 100.

	$\varepsilon^*_{\Delta C \leq 0.05}$			$\varepsilon^*_{\mathrm{LSR} \leq 0.50}$			$\varepsilon^*_{\mathrm{LSR} \leq 0.10}$			$\varepsilon^*_{\mathrm{FPR} \leq 0.10}$		
Dataset	dd	fixed	wid.	dd	fixed	wid.	dd	fixed	wid.	dd	fixed	wid.
lung	7	7	10	60	60	100	> 1000	> 1000	> 1000	—	250	—
pbc	30	60	60	60	250	250	1000	> 1000	1000	250	1000	1000
colon	10	10	10	30	30	30	1000	1000	1000	1000	1000	1000
rotterdam	10	30	30	10	10	10	1000	250	1000	60	60	60
flchain	5	5	15	30	60	60	100	100	250	> 1000	> 1000	> 1000

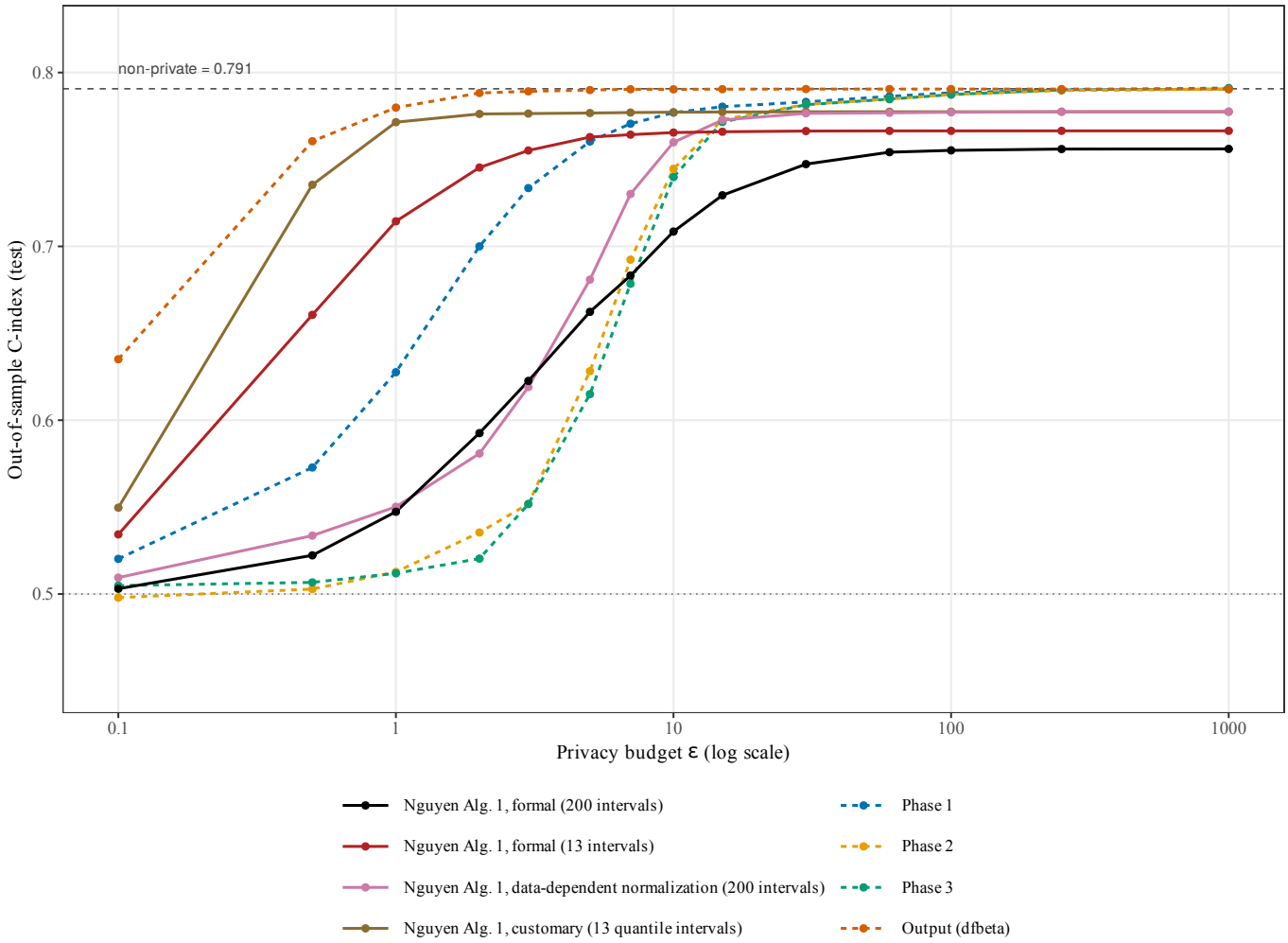


Fig. S6. Test C-index against ε on flchain for Algorithm 1 of Nguyên and Hui in four specifications (solid lines) and for Phases 1–3 and dfbeta output perturbation of this study (dashed lines; nominal ε). Formal (200 intervals) and formal (13 intervals): equal-width intervals on the fixed 5,223-day horizon and covariate normalization from the fixed box of the source protocol (Section S0); hence, the guarantee holds end to end. Data-dependent normalization (200 intervals): the same grid and time normalization, covariate normalization from the training data; no guarantee. Customary (13 quantile intervals): time and covariate normalization from the training data; no guarantee. $\Lambda = 10^{-1}$, published by the original authors, is applied unchanged in all four. Numerical values are listed in Table VI of the main manuscript.

S6. Hazard Ratios

Figs. S7 and S8 present the hazard-ratio distributions and the signed relative bias under Phase 1 that Section IV-F of the main manuscript summarizes; the relative bias is defined in Section III-G of the main manuscript.

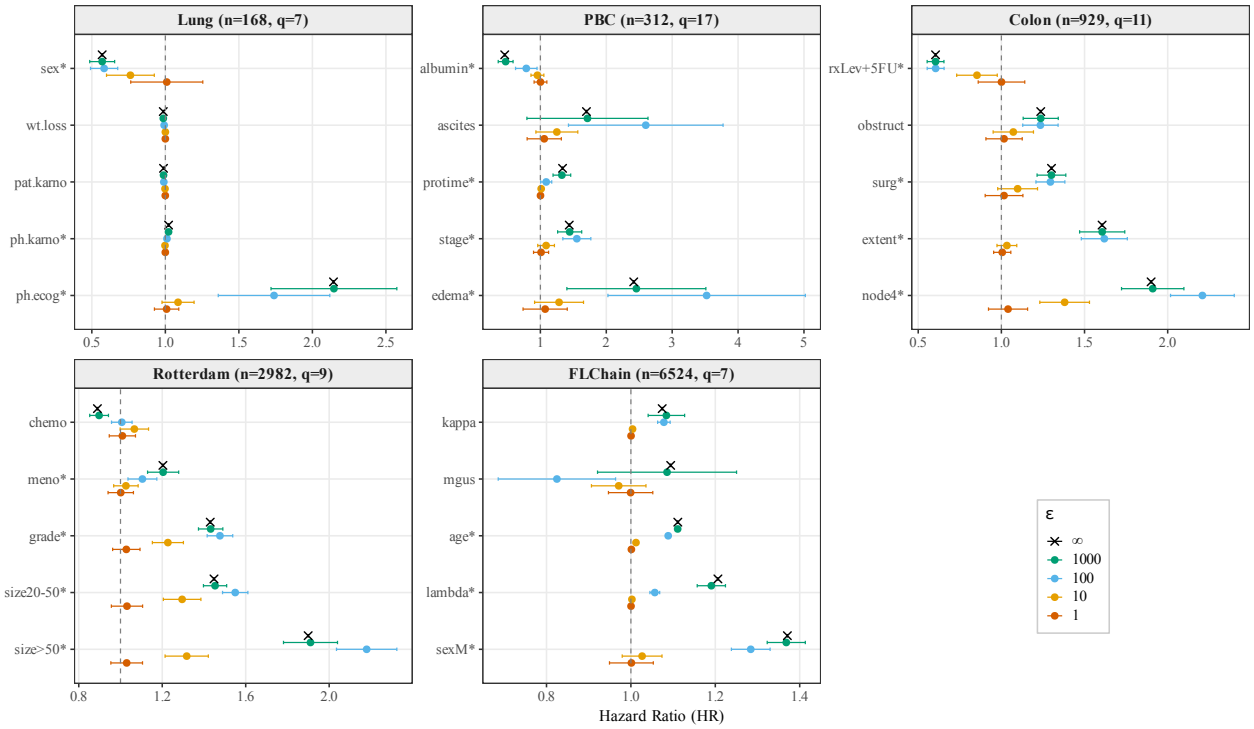


Fig. S7. Hazard-ratio distributions for the five leading covariates of each dataset under Phase 1. Dots are the mean over $B = 1{,}000$ iterations, error bars ± 1 SD, and $\times$ marks the baseline value at $\varepsilon = \infty$. At low ε every hazard ratio collapses toward 1.0, so that not even the direction of an effect can be read off. Variables are ordered by descending baseline coefficient.

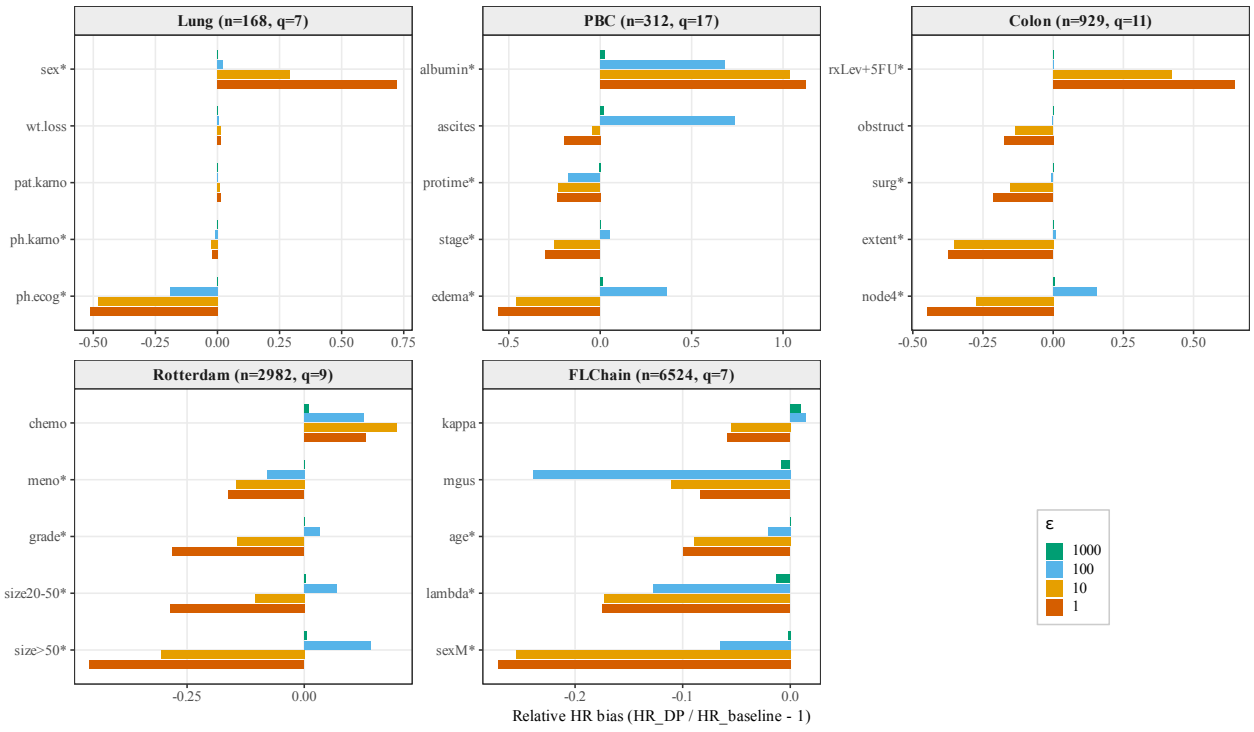


Fig. S8. Signed relative bias of the hazard ratio under Phase 1, $\mathrm{HR}_{\mathrm{DP}}/\mathrm{HR}_{\mathrm{baseline}} - 1$ as defined in Section III-G of the main manuscript (bars to the left of zero are negative biases), at $\varepsilon \in \{1, 10, 100, 1000\}$; $*$ marks coefficients significant at baseline, and the order matches Fig. S7. Since $\mathrm{HR} = \exp(\hat{\beta})$, this measure diverges under output perturbation at small ε and is reported for Phase 1 only.